



Multi-criteria and CNN analysis of Al₂O₃/TiO₂/egg shell/ATH ceramic fillers in glass fiber-epoxy composites

H. Mohit^a, V.V. Vamsi Krishna^b, M.R. Sanjay^{c,*}, Suchart Siengchin^c, Ibrahim M. Alarifi^d, Adel Alblawi^e, Mohammed A. Alamir^{f,***}, Femiana Gapsari^{g,**}

^a Department of Mechanical Engineering, Alliance School of Applied Engineering, Alliance University, Bengaluru, 562106, India

^b Department of Mechanical Engineering, JNTU, Pulivendula, Anantapur, Andhra Pradesh, India

^c Natural Composite Research Group Lab, Department of Materials and Production Engineering, The Sirindhorn International Thai German Graduate School of Engineering, King Mongkut's University of Technology North Bangkok, 10800, Thailand

^d Department of Mechanical and Industrial Engineering, College of Engineering, Majmaah University, Riyadh, Saudi Arabia

^e Mechanical Engineering Department, College of Engineering, Shaqra University, Dawadmi, Saudi Arabia

^f Department of Mechanical Engineering, College of Engineering, Jazan University, Jazan, Saudi Arabia

^g Mechanical Engineering Department, Faculty of Engineering, Brawijaya University, MT Haryono 167, Malang, 65145, Indonesia

ARTICLE INFO

Handling editor: P Rios

Keywords:

Egg shell
Inorganic fillers
Glass fiber
Epoxy
Hybrid composite

ABSTRACT

The increasing global demand for sustainable and high-performance composite materials has driven research into novel filler combinations that enhance thermal and mechanical properties while minimizing environmental impact. This study explores the synergistic effect of egg shell waste—a calcium-rich biofiller—combined with inorganic ceramic fillers (alumina, titania, and aluminum trihydrate) in glass fiber-reinforced epoxy hybrid composites. Unlike prior studies focusing on individual fillers, this unique blend offers multifunctional enhancements. The composites' physical, mechanical, and thermal properties were systematically analyzed across various filler proportions, revealing substantial performance improvements. The optimal composite (L7GEA-TAER) exhibited a density increase to 3015.84 kg/m³ and a 25.4 % reduction in water absorption. Mechanical tests demonstrated significant enhancements, with flexural and tensile strengths increasing by 2.14 and 1.57 times, respectively, while the glass transition temperature improved by 8.92 %. These enhancements are attributed to superior interfacial bonding, reduced polymer chain mobility, and uniform stress distribution. Advanced analytical techniques, including deep learning-based convolutional neural network (CNN) analysis of SEM microstructures, confirmed refined filler dispersion and improved structural integrity. This research highlights the potential of integrating biowaste with inorganic fillers to develop high-performance, sustainable composites suitable for demanding aerospace, automotive, and construction applications. The findings provide a comprehensive framework for optimizing filler combinations to meet specific industrial requirements while promoting circular economy principles.

1. Introduction

Rising societal problems and environmental awareness, encompassing the unsuitable utilization of petroleum products, and present environmental laws, which compel the current world to pioneer novel environmental-friendly constituents. Plant fiber is one among the diverse nature-friendly components. In addition, the plant fiber incorporated polymer laminates are more advantageous over synthetic

fabrics for their higher mechanical characteristics, lightweight, enhanced surface finish of molded parts laminate, available easily, renewable sources, minimum health hazard, biodegradable in nature and lower price [1–5]. Majority of the present industrial and academic research efforts are streamlined towards pursuing environment-friendly and biodegradable laminates comprising different plant fibers as reinforcement with their possible execution in different industrial and societal regions. These plant fiber laminates acquires normal to

* Corresponding author.

** Corresponding author.

*** Corresponding author.

E-mail addresses: mavinkere.r.s@op.kmutnb.ac.th (M.R. Sanjay), malamir@jazanu.edu.sa (M.A. Alamir), memi_kencrut@ub.ac.id (F. Gapsari).

<https://doi.org/10.1016/j.jmrt.2025.03.190>

Received 21 January 2025; Received in revised form 17 March 2025; Accepted 20 March 2025

Available online 21 March 2025

2238-7854/© 2025 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

remarkable physical and mechanical properties [6–8]. Furthermore, plant fibers have many constraints, because these fibers contain pectin, lignin, hemicellulose, waxy and cellulose components, which is liable for the absorption of moisture from the environment, tends to produce a weaker bond within the polymer and fiber. Additionally, as a result of the deviation in chemical structures of polymers and fiber, the difficulty remains same in coupling within polymer and plant fiber [9–12]. These are the causes behind the ineffective stress transfer during the interaction of the developed laminates and it can be solved with the addition of inorganic fillers within the matrix. It is typical to reinforce other components such as fillers, particles, fibers, whiskers etc., within the reinforcement materials to improve different characteristics such as physical and mechanical properties of the target laminates [13–17]. Usually, the effectiveness of adding fillers in laminates is observed to be inverse relationship to the size, whereas positively correlated to the filler surface area to volume proportion [18–20]. Currently, nanofillers are evolving as an outstanding fiber which can efficiently elevate and adjust the physical and mechanical characteristics of polymer laminates [21–24]. The higher surface area of the nanoparticles assists better interfacial bonding within the polymeric matrices which promote different characteristics of the laminates [25,26]. Plant fiber laminates incorporated with nano fillers are more sustainable, with enhanced mechanical characteristics, and decreased water absorption capabilities, and hence are widely integrated in construction sectors for railway coaches, ceiling paneling, consumer products, aerospace packaging, partition boards, etc. [27,28]. Moreover, when specific nano fillers are reinforced in the plant fiber polymer laminates, which may tend to higher hydrophobic formulations, UV protection, anti-odor, and anti-bacterial characteristics [29–31]. Presently, there are range of novel nanotechnology-based textiles in utilization in different industries such as clothing lines, space technology, beauty, and sports good.

The plant fiber addition in thermoplastic and thermoset laminates were investigated by Mohammed et al. [4] to examine how different types of chemical surface modifications influenced their tribological, water absorption, thermal, flame retardancy, and mechanical characteristics. The investigation determined that chemical surface modification on the plant fiber amplified adhesion between the polymeric matrices and surface of fiber that leads to enhanced the thermo-mechanical and physico-mechanical characteristics of plant fiber polymer laminates. A combined impact of the investigation on many bast fibers such as ramie, jute, hemp, banana, flax, and kenaf reinforced polymer laminates, encompassing the effect of chemical surface modification of their mechano-physical properties and their potential utilization was examined other researchers [32,33]. The SiO₂ nanoparticles was examined by Sinha et al. [6] as a filler to enhance the dynamic characteristics of epoxy laminates added with plant fibers. In this research, the dynamic and mechanical characteristics were studied with and without the reinforcement of nanoparticles and found that the addition of SiO₂ fillers improved the mechanical characteristics and also observed as lightweight, lower price and extraordinary qualities. Pujar et al. [34] investigated the influences of main reinforcements such as Kevlar, carbon, and glass fabrics, where as the auxiliary reinforcement such as CaO, TiC, TiO₂, MgO, Al₂O₃, graphite, etc. on the wear and mechanical properties of polymer laminates. In addition to different volume proportions and particle sizes of auxiliary reinforcement, the effect of main reinforcement's shape is also emphasized. The present issues and challenges with wear and mechanical properties of plant fiber reinforced thermoset laminates have been investigated by other researchers [28,35]. They revealed that the interfacial adhesion is primary factor for the wear and mechanical characteristics improvement. The frictional and wear characteristics of thermoset laminates based on natural fibers can be administered by selecting the orientation and appropriate quantities of fillers for the reinforcement.

Eventhough plant fiber reinforced polymer laminates have many benefits, containing lightweight with higher wear resistance, excellent physical, mechanical and thermal characteristics, their optimal

utilization in generic applications still acquires developing accurate numerical modelling, prediction technique, and innovative design. Producers frequently utilize a trial and error technique by modifying design factors to avoid mistakes. Also, interdependent and conflicting performance condition was generally assessed when choosing polymeric laminate designs. Present investigations have observed the deployment of the multi-criteria decision making (MCDM) techniques are essential to evaluate the optimal polymer laminate which can encounter a range of performance condition to demonstrate the challenges in the selection of entire composite design [36–40]. Many MCDM techniques such as MOORA, COPRAS, WASPAS, SAW, etc. have been applied successfully to design polymer laminates [41–44]. Although many MCDM techniques were accessible in the recent literature to support researchers in opting the best laminate, it was detected that the optional ranking was influences from the condition weight in certain techniques, whereas others were fairly challenging to recognize and complicated to utilize, demanding comprehensive mathematical information. Hence, a straightforward, systematic, and rational technique is yet required to prevail the laminate selection problems. The selection criteria method is more particular to perceive than another MCDM technique due to no evaluation of the related significance of the conditions and the entire value is examined by applying the statistical understanding [45–47].

A systematic frameworks in the sector of material science and engineering, especially data-driven, theoretical, empirical and computation. Among these models data-driven meta-architecture combines the other three patterns and is intensely becoming favored as huge substances data from simulations or experiments develops. In principle, the data-driven framework of science initiates the novel sector of materials information science that has insightful effect on illuminating structure-characteristic bond [48]. The information systems methods provide increment to substitute machine learning techniques which can comprehend precise prediction using previous information rather than simulations or experimentations [49–51]. This technology is generally comprised of three different procedures such as mapping between output and input parameters, attainment of efficient historical information, and statistical evaluation of data-intensive structures. The dataset collected from input is generally obtained from original images or conventional models whereas the output data can be interpreted with simulations or experimentation results [52,53]. To collect the statistical presentation of the material arrangement, distinct techniques have been generated such as two-point spatial correlation, pair correlation formulation, K function, and nearest neighbor distance [54,55]. Among these techniques, the two-point correlation have been substantiated to be more effective method to attain this [56]. Unsupervised learning techniques can be used for complexity decrement of the fingerprint attributes such as a successful trial is principal constituent examination [56,57]. Conclusively, the networks between characteristics and fingerprints of attention are interpreted by supervised learning techniques like main component regression, artificial neural network, etc. [58]. Among these regression methodologies, deep learning method like convolutional neural network because of the exceptional performance in visual data processing [51,52,59] are also widely applied for pursuing material characteristic relationships of substances.

Furthermore, till now no investigation has been observed on the utilization of eggshell, and inorganic ceramic reinforcements as efficient fillers to enhance the physical, mechanical, and thermal characteristics of epoxy polymer hybrid composites using MCDM and CNN techniques. The addition of fillers in the polymer composites handles crucial limitations in conventional fiber-reinforced materials through many principles. Inorganic particles such as alumina, titania, and aluminum trihydrate are specifically advantageous as they improve interfacial adhesion within fillers and matrix through mechanical interlocking and chemical bonding, thereby assisting more efficient stress transfer. These fillers operate by engaging the interstitial gaps in the polymeric matrices, limiting chain mobility and incrementing the composite stiffness. Additionally, ceramic particles contribute to enhanced thermal

stability by producing thermal barriers and decreasing heat transfer rates within the structure of composites. The surface functionality of inorganic particle can be customized to produce hydrogen or covalent bonds within the polymeric matrices, leads to improved transfer effectiveness and decreased internal damages. The egg-shell waste supplement these impacts by offering calcium-rich fillers which enhance crack-propagation resistance while simultaneously handling environmental concerns through waste recognition. The synergistic consolidation of multiple particles produces a multi-scale reinforcement scheme where larger fillers enhance impact resistance while nanofillers improve matrix characteristics under the molecular level [60–65]. The strategic reinforcement of multiple fillers in glass-fiber reinforced polymer composites focuses on many crucial restriction inherent to single-filler schemes. Recent investigations have illustrated that individual fillers offer particular benefits, they often fails to simultaneously improve multiple characteristics required for high-performance applications [24]. For instance, single nano-particles produce localized characteristic enhancements but struggle to offer balanced improvement throughout the matrix of composite [19,21]. This multi-scale reinforcement technique provides concurrent focusing of multiple characteristic improvement [59,66–68]. However, comprising ceramic fillers with polymeric materials produces a complementary impact where inorganic constituents enhance mechanical strength and thermal stability where polymeric resin improve processing and flexibility properties [7]. The reinforcement of waste-derived particles along with traditional inorganic fillers presents an innovative technique to attaining multifunctional characteristic improvement where its simultaneously tackles the environmental sustainability issues through waste valorization [29–31,69,70]. Hence, in the current research investigation, eggshell, alumina (Al_2O_3), titania (TiO_2), and aluminum trihydrate (ATH) reinforced epoxy hybrid laminates manufactured and experimented for their physical (density, porosity, contact angle, and water absorption), mechanical (tensile, compression, flexural, impact, interlaminar shear strength, and shore D hardness), and thermal stability characteristics have been studied. Selecting the best laminate was tough since each laminate implemented differently for the analyzed characteristics [71]. As an outcome, in this investigation, the MCDM technique is illustrated to choose the best laminate with the highest degree of fulfillment for all analyzed characteristics. This work also aims to demonstrate and explore the effectiveness of utilizing materials information framework to extract the structure-characteristics relations of epoxy hybrid composites with defects, that has never been done before. The accuracy of the developed information is quantitatively and qualitatively confirmed by comparing with experimental outcomes [68]. Then, the fracture surface were quantified by utilizing two-point spatial relations. Eventually, the robust structure, and highly efficient property connections were traced through back propagation neural network technique. The influences of image resolution and dataset size on the precision of the final outcomes were also examined.

2. Materials and methods

2.1. Materials

The eggshells of gallus domesticus (also known as country chicken) were collected from poultry farms located in Putaparthi, Andhra Pradesh, India. The Al_2O_3 (150 μm), TiO_2 (150 μm), and ATH (150 μm) were procured from Carborundum Universal, Chennai, India in the 99 % of purest form. Synthetic epoxy LY 556 with hardener HY 951 was applied as the matrix material and obtained from Powertek stones and chemical solutions, Bengaluru, India.

2.2. Processing of fillers

The obtained eggshell was cleaned with fresh tap water to remove the dirt particles and dried in sunlight for 2 days under atmospheric

pressure and temperature. The dried eggshell was converted into microparticles using mortar/pestle and sieved in micro-mesh of 150 μm (average size) to get the eggshell micro-fillers. Then, the micro-fillers of eggshell kept in industrial furnace under 100 $^\circ\text{C}$ to remove the residual moisture contents. Afterwards, the eggshell micro-fillers were combined with inorganic particles such as Al_2O_3 , TiO_2 , and ATH in the combination as EA (Eggshell with Al_2O_3), EAT (Eggshell with Al_2O_3 and TiO_2), and EATA (Eggshell with Al_2O_3 , TiO_2 , and ATH) using high energy planetary ball mill separately for 5 h with interval time of 20 min for every 1 h. The ball milling was performed under 200–250 rpm with balls to powder weight ratio of 12:1 with zirconia ceramic balls. The processed micro-particles such as EA, EAT, and EATA were kept separately and applied as reinforcement to the epoxy matrix.

2.3. Fabrication of epoxy hybrid composite

The processed fillers such as EA, EAT and EATA were reinforced into the epoxy polymer through a controlled dispersion technique. Initially, the epoxy resin was preheated to 40 ± 2 $^\circ\text{C}$ for 2 h to decrease viscosity and facilitate uniform filler distribution. The fillers were gradually incorporated to the polymer while controlling continuous mechanical stirring at 380 rpm for 2 h using a digital overhead stirrer with temperature control. To remove entrapped air bubbles during mixing, the mixture was transferred to vacuum degassing at -0.9 bar for 30 min under room temperature (28 ± 2 $^\circ\text{C}$). The curing the composite, 10 wt% of hardener was incorporated to the resin mixture and stirred vigorously for 30 min using a glass rod, followed by a second degassing cycle (10 min under -0.8 bar) to eliminate any air entrapped during the process of hardener addition. The glass fabric reinforcement was prepared by sectioning 7 layers of E-glass fabric into 300×300 mm and conditioning under 50 ± 2 $^\circ\text{C}$ for 3 h to eliminate any moisture prior to lamination. The wetlayup technique was used for composite manufacturing where the degassed resin mixture was applied over the glass fabric layer (which is placed on steel plate die) using a brush and roller to ensure homogeneous dispersion and appropriate impregnation, which is followed for next 6 layers in stacking sequence and orientation maintained at 0° to assure directional characteristic consistency. The glass-fabric reinforced epoxy hybrid composite was then compressed between the steel plates under room temperature with a load of 15 kg (0.42 MPa as equivalent pressure) over the top surface and kept aside for 24 h to permit for complete room-temperature curing. After the initial curing, the composite was removed from the steel mold and transferred to heating furnace with temperature of 70 ± 2 $^\circ\text{C}$ at a rate of 2 $^\circ\text{C}/\text{min}$ controlled at this temperature for 48 h and slowly cooled to a room temperature at a rate of 1 $^\circ\text{C}/\text{min}$ to minimize the residual thermal stresses. This post-curing method was implemented to optimize the crosslinking density of the epoxy polymer and improve the interfacial bonding within the fillers, fabric and matrix [72]. The fabrication details of the different epoxy hybrid composites are illustrated in Table 1.

2.4. Composite characterization

2.4.1. Density

The actual density (d_a) of the laminates was investigated as per ASTM D 792-20 by a soaking technique and the evaluation was performed in deionized water using ALN balance, AXIS. The weighing balance was calibrated using certified standard weights with ± 0.001 g accuracy. The balance was verified before every testing from standard weights. The theoretical density was measured according to equation (1).

$$d_t = \frac{d_m \times d_f}{W_e \times d_f + W_f \times d_m} \quad (1)$$

where d_t , d_m , d_f , W_e , and W_f are theoretical density, density of matrix, density of filler, weight of epoxy polymer, and weight of reinforced

Table 1

Fabrication details of Al₂O₃/TiO₂/egg shell/ATH/glass fabric reinforced epoxy hybrid composites.

Composite Type	Description	Egg shell	Al ₂ O ₃	TiO ₂	ATH	Glass Fabric
L7GER	Glass fabric reinforced with epoxy resin	–	–	–	–	7 layers
L7GEER	Egg shell fillers and glass fabric reinforced with epoxy resin	9 wt %	–	–	–	
L7GEAER	Egg shell, alumina fillers, and glass fabric reinforced with epoxy resin	4.5 wt%	4.5 wt%	–	–	
L7GEATER	Egg shell, alumina, titania fillers, and glass fabric reinforced with epoxy resin	3 wt %	3 wt %	3 wt %	–	
L7GEATAER	Egg shell, alumina, titania, aluminum trihydrate fillers, and glass fabric reinforced with epoxy resin	2.25 wt%	2.25 wt%	2.25 wt%	2.25 wt%	

filler. The theoretical density of eggshell powder was considered as 2.35 g/cm³ based on the published literature for calcium carbonate which is the primary constituent of eggshell, containing 95 % of its composition. This quantity was verified from different literature sources and applied in the rule of mixtures to calculate the theoretical density of the composite materials [73]. To measure the void concentration of the epoxy hybrid composites, equation (2) was employed [74].

$$V_c = \frac{d_t - d_a}{d_t} \quad (2)$$

2.4.2. Porosity

The porosity evaluation of the assessments was conducted on samples which obtained from the mild steel uni-directional plate. In this test, the samples were polished, and internal structure of the laminate was captured in the form of two-dimensional pictures to measure the surface proportion of the gaps using optical microscope, Evident Microscope. The equipment utilized for porosity measurement was calibrated using a certified stage micrometer with 0.01 mm graduations. Magnification accuracy was verified to be within ± 1 % across the complete field of view. The image calibration was conducted using ImageJ software with traceable measurement standards. The resolution of system was 0.5 μ m at 300 $\times$ magnification.

2.4.3. Contact angle

The OCA 15 LJ Contact angle meter, Data Physics Instrument device was equipped to determine the contact angle of the epoxy hybrid laminates. To calculate the contact angle, droplets of water were poured at different portions of the specimen and a mean of 12–15 measurements were considered as a value. The equipment was calibrated using optical calibration technique and reference liquids with known surface tensions. The accuracy of measurement was controlled under $\pm 0.1^\circ$ for contact angle readings.

2.4.4. Water absorption

The water absorption characteristics of the laminates were investigated according to ASTM D 570-22. The specimen were sectioned deliberately into a chip of 20 $\times$ 20 $\times$ 5 mm before the analysis. The laminate specimen were soaked in deionized water for 90 days and modify in weight is estimated.

2.4.5. Mechanical properties

The tensile experiment was performed as per the ASTM D 638-14 standard using an Universal Testing Machine, Instron of 100 kN grip capability. The dimensions of the sample was 100 $\times$ 15 $\times$ 5 mm in a dog bone structure with 10 mm of gauge length and crosshead speed of 2 mm/min. The longitudinal compression testing of laminate was conducted from 100 kN load cell and crosshead speed of 2 mm/min as per the ASTM D 695-23 with sample dimensions of 25 $\times$ 30 $\times$ 5 mm. Whereas the bending test was performed as per the 3-point flexural medium as per the ASTM D 790-17 with crosshead speed of 2 mm/min, dimensions of sample as 80 $\times$ 12.5 $\times$ 5 mm, and 60 mm of support span. Interlaminar shear strength (ILSS) of epoxy hybrid laminates were experimented which depended on small beam length test according to the ASTM D 2344-22. The test was conducted using 5 kN of loading capability and data encoder connected at cross head speed of 2 mm/min. The load cell of 100 kN of Universal testing machine was calibrated using ASTM E4-21 protocols with precision of ± 0.5 % of the indicated load. The displacement were measured with ± 0.01 mm.

The izod impact experiments were carried out as per the ASTM D 256-24 using an Impact tester, Zwick Roel equipment with 20 J Hammer. The unnotched samples were produced from composites in 62.5 $\times$ 11 $\times$ 5 mm as per standard. The absorbed energy quantification and failure mediums were collected under the constant energy of 20 J. The impact tester equipment was calibrated as per the ISO 13802 using reference pendulum and certified calibration weights. The energy scale was verified to a precision of ± 0.5 % of the full scale (20 J). Angular position sensors were calibrated to $\pm 0.1^\circ$ to assure accuracy measurement of impact energy absorption. The instrument was verified monthly using standard test samples with known impact characteristics.

The same sample of the tensile test is applied to determine the shore D hardness of the laminate using Indenter, Genex. The equipment was calibrated using reference hardness blocks covering the range between 30 and 90 shore D units with measurement accuracy of ± 0.01 shore D unit. Every sample was tested under 10 different positions with shore hardness tester at room temperature. Ten samples were experimented for each mechanical test, with an allowable error margin of within 5 %.

2.4.6. Thermal stability

The thermogravimetric analysis (TGA) was carried out with TGA 8000 thermogravimetric analyzer, PerkinElmer between the temperature ranges from 30 to 600 $^\circ$ C, with a heating rate of 5 $^\circ$ C/min in nitrogen atmosphere. Specifically, during these tests, specimen with weight of approximately 15 mg were positioned in alumina crucible of 85 μ L, which offers adequate space for adequate gas evolution and thermal decomposition [75–77] and the total flow rate was fixed as 60 mL/min. The gas flowrate of 60 mL/min was chosen to assure the complete elimination of oxygen from the system and avoid unwanted oxidation reactions during the evaluation that is crucial for accurate measurement of the degradation characteristics of these polymer composites [78–80]. The equipment calibrated monthly from reference materials such as zinc and indium with known thermal transition temperatures. The balance was calibrated using standard weights of 100 mg with accuracy of ± 1 $^\circ$ C and ± 0.1 % for temperature and weight measurement readings respectively.

Differential scanning calorimeter (DSC) was performed from DSC 4000, PerkinElmer machine in operating temperature between 30 and 300 $^\circ$ C at heating rate of 5 $^\circ$ C/min under nitrogen atmosphere, with a specimen of 15 mg in aluminum pans of 40 μ L capacity to assure optimum thermal contact between the pan and specimen for precise heat flow evaluation [81–83]. The calibration of equipment using pure zinc and indium standards for enthalpy of ± 2 % and temperature accuracy of ± 0.5 $^\circ$ C. The baseline calibration was conducted prior to testing to assure stability of measurement.

2.4.7. Fourier transform infra-red (FTIR) spectra

The presence of functional groups on epoxy hybrid composites was

affirmed from FTIR spectrum, FTIR spectrometer, PerkinElmer Spectrum 2. The FTIR sample was prepared with 95 wt% of potassium bromide and 5 wt% of fabricated epoxy composite was combined together using pestle/mortar to form a 1.5 mm of pellet. The prepared sample was kept in the holder and wavenumber between 4000 and 400 cm^{-1} with an interval of 1 cm^{-1} was measured with a scan rate of 32 scans/min. The wavelength calibration was conducted using a polystyrene reference thin film with existing absorption peaks. The resolution was controlled at 1 cm^{-1} and the system ensures an accuracy of wavelength as $\pm 0.1 \text{ cm}^{-1}$.

2.4.8. SEM

The microstructure of the fractured surfaces after tensile experiments were examined using an Axia ChemiSEM (Scanning Electron Microscope), Thermo Fisher Scientific under 10 kV. Before SEM evaluation, the fabricated laminates were sputter-coated with plasma gold ion for around 60 s using sputter coater device to form the conductive nature samples and to avoid charge accumulation from electron capture. The calibration was performed using gold on carbon resolution standard. The accuracy of magnification was controlled at $\pm 2 \%$ with capacity of

resolution as 1 nm under 15 kV.

2.5. Integrated deep learning, MCDM and statistical analysis

Since the developed epoxy hybrid composites have sufficient physical, mechanical, and thermal characteristics, the reinforcement of different inorganic fillers should be optimized in order to attain all essential objectives. In this investigation, advanced machine learning methodologies with MCDM and conventional statistical analysis approach is applied to choose the best epoxy hybrid composite. This comprehensive technique permits us to examine the complex relationships within the physical, mechanical, and thermal characteristics and optimize the multiple conditions in parallel.

2.5.1. Deep neural network (DNN) technique

A DNN technique with 3 hidden layers which contain 300 neurons for regression assessment. This method was selected because of its capability to obtain complex, non-linear relationships within input factors and target characteristics that is critical when modeling composites varied properties. Composites usually present complex

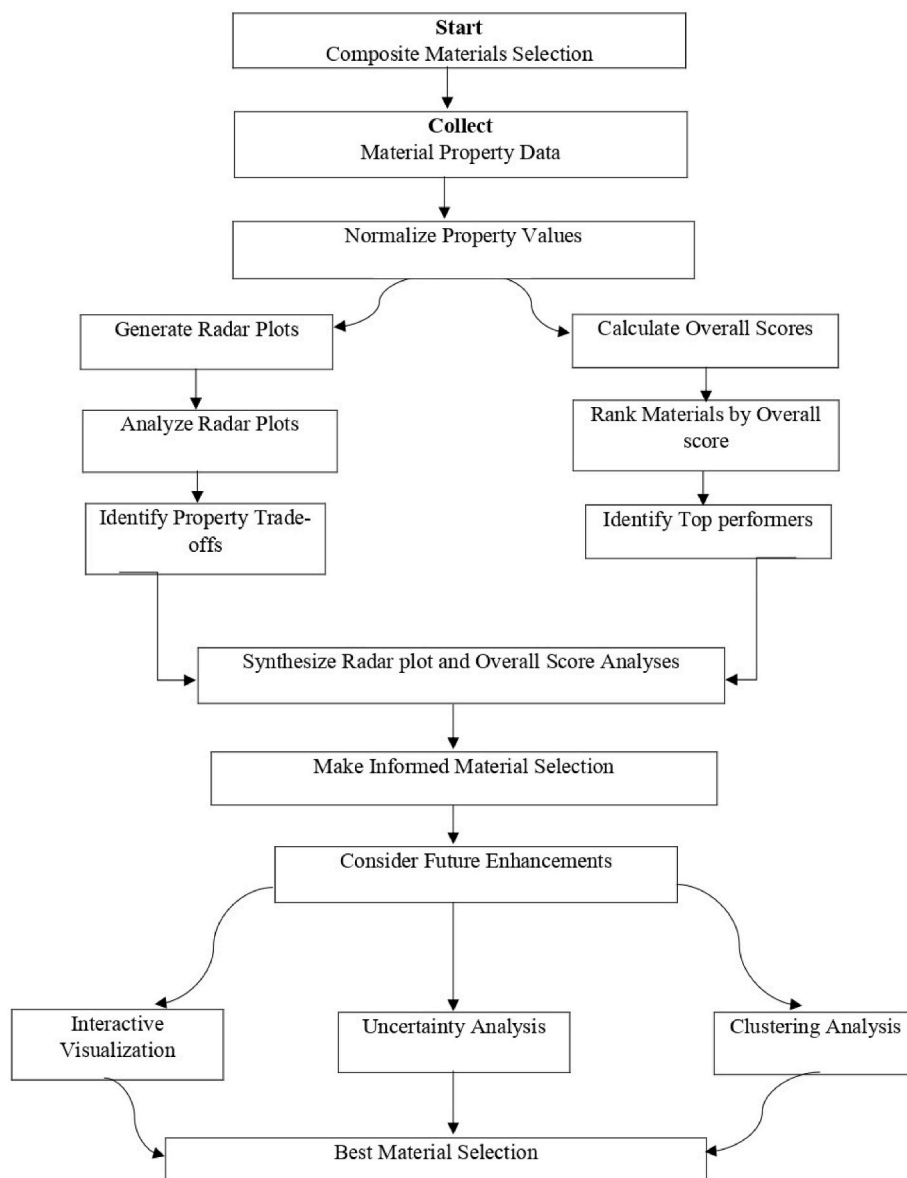


Fig. 1. Flow chart of composite material selection using MCDM and DNN technique.

interactions within their constituents, pioneering characteristics which are not readily predictable from linear models. The deep neural networks outperform at learning these rigid relationships, creating them unique for this purpose.

2.5.2. MCDM

The MCDM technique (Fig. 1) for selection of best composites includes a comprehensive normalization and scoring process. Initially, all materials characteristics should be normalized to a typical scale, generally between 0 and 1, to assure unbiased comparison across different characteristics with ranging magnitudes and units. For characteristic which should be maximized are tensile strength, young's modulus, compression strength, compression modulus, flexural strength, flexural modulus, impact strength, shore D hardness, glass transition, onset, endset temperature and carbon residue, equation (3) has been applied. Whereas the characteristics should be minimized are density, water absorption, porosity, and contact angle, equation (4) has been employed.

$$N_{max} = \frac{(V_a - V_{min})}{(V_{max} - V_{min})} \quad (3)$$

$$N_{min} = \frac{(V_{max} - V_a)}{(V_{max} - V_{min})} \quad (4)$$

where N_{max} , N_{min} , V_a , V_{max} , V_{min} are Normalized value for maximized property, Normalized value for minimized property, actual, maximum, and minimum value, respectively. This normalization process is critical as it permits to combine and compare diverse characteristics on balanced ground. After normalization, the overall score for each composites was estimated by comprising these normalized quantities. This combination procedure consider the contextual importance of every characteristic for the particular application, often implementing a weighted sum method where every normalized characteristic value is amplified with a predetermined weight parameter before compilation. This method originates from the complex behavior of epoxy hybrid composite selection that generally include handling multiple, periodically discordant properties, and criteria. By normalizing characteristics and estimating an overall score, materials could be unbiasedly evaluated across different parameters, assuring a complete analysis. This technique gathers the multifaceted behavior of material performance in a comparable criterion. However, modifying the weights in the scoring system permits for adapting the analysis to particular application requirements, focusing some characteristics over others as required. This flexibility produces the method adjustable to a broad range of composite selection schemes, based on the application where both strength and weight are important.

2.5.3. Performance and correlation analysis

In epoxy hybrid composite modeling, performance graphs assist as critical tools for optimizing and perception of deep learning model characteristics. Plotting mean absolute error (MAE), validation and training over training epochs offers significant investigative insights into the learning techniques and supports to determine possible problems like underfitting or overfitting. The overfitting is identified when validation loss increments and training loss reduces, whereas the underfitting is signified from steadily higher loss values for validation and training sets. Observing these graphs notifies decision about model scheme, duration of training, and hyper factors tends to a more generalizable and rigid model for forecasting the epoxy hybrid composite characteristics. Supplementing the performance examination, a comprehensive correlation assessment of different composite characteristics is performed. This assessment is visualized through an improved correlation heatmap within characteristics providing a complete perspective of interrelationships dataset. Observing these relationships is critical for material optimization and design, as it recognizes

characteristics which lead to range together or in adversity. Strong positive correlations signify characteristics that reduce or increment simultaneously where strong negative formulations recommend an reverse connection. Weak correlations could indicate to characteristics which can be changed autonomously. This visual illustration facilitates quick determination of essential correlations, navigating feature selection and understanding the basic chemistry and physics of epoxy hybrid composite. Utilizing these observations notifies decision-making in the composite design procedure, possibly revealing unforeseen relationships which could tend to processing method or new composite development.

2.5.4. Regression and error analysis

The methodology integrates comprehensive regression assessment to measure the performance of models in forecasting individual epoxy hybrid composites characteristics. For every characteristic, regression graphs are developed to perceptibly illustrate the relationship within actual vs predicted values of models. These curves offer inherent understanding of the accuracy of model across the total range of property quantities. Identically, points of these curves should closely complies a diagonal line, signifying a strong agreement within the actual and predicted quantities. Variations from this diagonal can disclose systematic biases in the predictive capacity of models for particular range of values. Adding these visual illustration, R-squared quantities were measured for every characteristics, which is also termed as determination coefficient, evaluates the variance ratio in the dependent parameter (the material characteristics) which is predictable from the independent parameters (the input factors). R-squared quantities range between 0 and 1 with values near to 1 signifying better predictive characteristics. This metric offers a numerical evaluation of the capacity of model to gather the underlying scheme in the data for every particular property, permitting for quantitative comparisons across model iterations and different characteristics. To improve the understanding the accuracy of predictive model, a dual approach error assessment method is considered, visualizing and estimating root mean squared error and mean absolute error for every predicted characteristics. Mean absolute error offers a mean of the absolute deviation within actual and predicted values, providing a direct explanation of the mean prediction error of model. It manages all errors equally, despite of their degree. Root mean squared errors values estimates the square root of the mean of squared variation within actual and predicted values. By adjusting the error before normalizing, root mean squared error provides more weight to huge errors, producing it specifically responsive to deviations or considerable forecasting flaws. These indicators are illustrated for every characteristic, generally using graphical presentation, facilitating for simple comparison among distinct characteristics. This dual signifier method offers a detailed view of the performance of model which contains mean absolute error provides a simple comprehensible mean error, while root mean squared error emphasizes any characteristics where the model intermittently produces huge errors. Concurrently, these indicators provide a comprehensive examination of the model's forecasting precision across the varied range of composite material characteristics, directing refinement of the model and notifying confidence in its forecasting for distinct kinds of characteristics.

2.6. Convolutional neural network

The assessment of scanning electron microscope (SEM) pictures was performed from convolutional neural network (CNN) technique. CNNs are specifically well-known for image examination assignments because of their capability to learn autonomously structured features from raw data. In this investigation, a custom CNN scheme adapted and developed for SEM micrographs evaluation (Fig. 2). The CNN was designed with multiple convolutional layers, each followed by batch normalization and ReLU activation functions. Max pooling layers were interspersed to reduce spatial dimensions and capture broader features. The network culminated in fully connected layers for final feature integration and

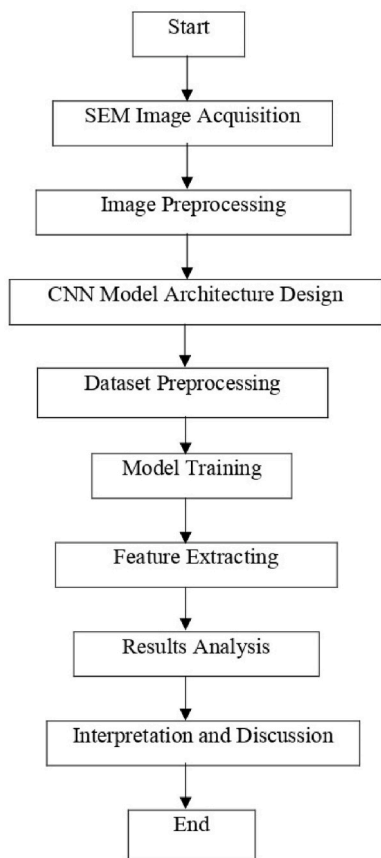


Fig. 2. Flow chart of SEM micrographs analysis using CNN model.

classification. To prevent overfitting, dropout layers were incorporated, and data augmentation techniques such as random rotations, flips, and brightness adjustments were applied to the training dataset [84]. The SEM micrographs were pre-treated to assure size standardization and varies the intensity of pixel. A dataset of labeled SEM micrographs was aggregated with professional comments for various characteristics of interest like patterns of fracture, and surface roughness. This dataset was divided into testing, training and validation with a proportion of 15:70:15, respectively. The CNN was trained using stochastic gradient decent (SGD) method with momentum, establishing a organizer to modifiable rate of learning during the training process. The cross-entropy loss was applied as the optimization condition. The training was performed for 100 epochs, with early halting employed based on the set of validation performance to avoid overfitting. After training, the CNN was used to an independent set of SEM micrographs to collect different quantitative attributes. These attributes contains roughness estimate, intensity standard deviation, average intensity, fracture estimate, and many Gray level Co-occurrence Matrix (GLCM) factors like homogeneity, energy, correlation, and contrast [84]. Also morphological characteristics such as circularity, perimeter, and average object are calculated. Frequency domain assessment was conducted from Fast Fourier Transform (FFT) to measure FTT energy. Statistical calculates consisting of entropy, kurtosis, and skewness were also inferred from the microstructure histograms.

2.7. Statistical analysis

In the current study one-way ANOVA (Analysis of Variance) were conducted from JMP Pro 17 software. Mean quantities were employed as statistically equivalent if p value is lower than 0.05. From this condition, there is chance of 5 % which states that two characteristics were examined similar and also statistically deviated.

3. Results and discussion

3.1. Effect on density

Information of density and void concentration in a specific laminates is necessary to evaluate the quality. The actual, theoretical, and void proportion of laminates were calculated as per the particular standards and the outcomes were depicted in Table 2. The theoretical density of laminates increases with the addition of new inorganic fillers. Similar pattern is also examined in actual density of the hybrid laminates. The actual densities collected were lower than the theoretical quantities and incremented with addition of new inorganic filler content. The sample L7GER exhibits that there is no filler reinforcement which is considered as the reference part to interpret the increment trend in densities with the reinforcement of E, EA, EAT, and EATA filler contents. The improvement in density of 2035.61, 2320.59, 2645.49, and 3015.84 kg/m³ for L7GEER, L7GEAER, L7GEATER, and L7GEATAER composites, respectively. The sample L7GEATAER produces lower voids that are suitable to produce laminate, though it is applied in the investigation to realize the condition of EATA filler content. There is a considerable deviation in theoretical and actual densities of the epoxy hybrid composites that can be correlated to the void concentration (Table 2). These voids are usually created because of entrapment of air bubbles in the matrix during the manufacturing of laminates [66]. The observation inferences that addition of multiple fillers considerably increments the void concentration in the laminates that mainly affects other physical characteristics [67,85].

3.2. Effect on porosity

The porosity was measured for every laminate, employing the reliance of the void production on stacking sequence and preformed material. The porosity characterization through microscopic visualization offers the best characteristics over the thickness of the laminate. Hence, the porosity evaluation was performed from optical microscope through the laminates thickness with 300× magnification and ImageJ software. The porosity content of L7GEATAER sample was 3.03 % where the L7GER exhibited a porosity of 3.73 % as shown in Fig. 3 (a), due to the EATA fillers showed a higher range of impregnation of fillers within matrix [69,70]. As predicted, the other epoxy hybrid composites such as L7GEER, L7GEAER, and L7GEATER offered transitional porosity outcomes with values of 3.54 %, 3.36 %, and 3.19 %, respectively. The reinforcement of E, EA, and EAT fillers was liable for the lower porosity content of the epoxy hybrid composite, and the lower variance that is assigned to more uniform impregnation, specifically for stacking sequence scheme [86].

3.3. Effect on water absorption

The water absorption characteristics of the epoxy hybrid composites samples were investigated up to 90 days. The water absorption characteristics of each sample is shown in Fig. 3 (a). In initial stage of the experiment, the water absorption capacity increments swiftly, proceeded by a deceleration and then reached to steady state phase [87]. The weight gain is maximum for reference sample and is 4.25 %.

Table 2

Density and void content of Al₂O₃/TiO₂/egg shell/ATH reinforced glass fabric-epoxy hybrid composite.

	Actual Density	Theoretical Density	Void Content
L7GER	1785.62	1839.19	2.91
L7GEER	2035.61	2086.50	2.44
L7GEAER	2320.59	2401.81	3.38
L7GEATER	2645.47	2730.13	3.10
L7GEATAER	3015.84	3085.24	2.25

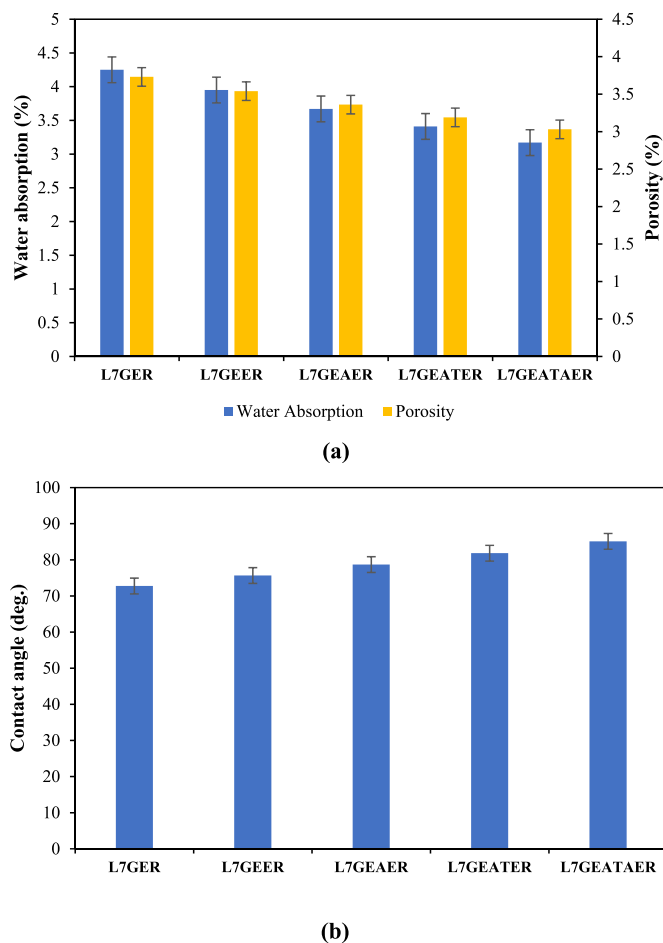


Fig. 3. Physical properties of Al₂O₃/TiO₂/egg shell/ATH reinforced glass fabric-epoxy hybrid composites (a) Water absorption and porosity, (b) Contact angle.

However for L7GEER, and L7GEAER is more than 3.5 %, because of the existence of fillers and fabrics in the epoxy resin. The void developed at the matrix-filler-fabric interface, hollow structure of fillers, fracture creation in the polymeric matrix, and hydroxyl functional groups existence in the fillers affect the water intake of the laminates [87–90]. After the addition of EAT and EATA fillers, the water absorption capacity decreases up to 3.41 %, and 3.17 %, respectively that may be because of the production of the strong bonding within the matrix, and hence water particles restricts and water absorption decreases. The production of microcracks and the distortion of egg shell fillers produced new way for water movement and thus higher water absorption capacity [91–93].

3.4. Effect on contact angle

The contact angle was determined to identify whether the fabricated epoxy hybrid composite is hydrophobic or hydrophilic. It provides details on the wettability and surface characteristics of the polymer. The super-hydrophilic and super-hydrophobic surface states that the contact lower than 10° and higher than 90°, respectively. The composites with decreases in water absorption capacity, and higher wettability with better self-cleaning capability is termed as super-hydrophobic and super-hydrophilic [30,94–96]. For outdoor use, the composite schemes are generally covered with enamel paints and recommended in hydrophilic surfaces. The analysis of the contact angle evaluation of reference samples and hybrid laminates is presented in Fig. 3 (b). The reference sample exhibits a contact angle of 72.75°, conventional for a hydrophilic component, and moisture content can quickly disperse over it.

Furthermore, the epoxy hybrid laminates exhibit increased contact angle, and the highest contact angle was examined for L7GEATAER composite even though it possesses hydrophilic phase. This exhibits that the engagement between the epoxy and water particles decreased in epoxy hybrid composites when compared with reference sample. The contact angle mainly relies on the roughness of the laminates. The surface of reference sample is smooth, whereas the surface of hybrid laminates is rough because of the existence of mixed fillers at the laminate surface which is also the cause for increment in contact angle of the laminates. The contact angle of the reference sample without filler (L7GER) measured 72.75°, slightly lower than pure epoxy's value of 75.6°. The L7GEATAER composite showed higher contact angle because of their individual fillers having lower contact angles (TiO₂: 30–50°, Al₂O₃: 62.6°, ATH: 58.1°) and their combination with higher contact angle materials like E-glass fiber (84°) and eggshell (80°) in the matrix which produced composite with improved hydrophobicity. This increment in contact angle can be assigned to surface roughness produced by the mixed particles at the composite surface that decreased the interaction between water molecules and epoxy when compared with the reference sample [97–107].

3.5. Effect on tensile characteristics

The structural tensile experiment is broadly applied to offer fundamental design data on the stress of materials and as an approval trial for the composite specifications. The quantities of ultimate tensile strength, elongation at break, and young's modulus are measured from the stress-strain graphs shown in Fig. 4 (a) obtained during the tensile experiment. The influence of without and with fillers (E, EA, EAT, and EATA) on the average values of tensile strength, elongation at break young's modulus, and yield strength of the epoxy hybrid laminates are shown in Fig. 4(b) and (c). The reinforcement of E, EA, EAT, and EATA fillers to the matrix incremented the strength of the laminate with respect to reference sample. The source of these characteristics may be assigned to the enhanced stress transfer capability to the matrix with an increment in addition of multiple inorganic fillers [108]. Nonetheless, the young's modulus quantities are examined to be highest for EATA filler reinforcement because it generated good resistance to divergence. Many present investigation on polymer laminates have also examined the inequality in attaining the highest quantities for tensile strength and modulus under different types of filler reinforcements [14,109–111]. The assigned effect of the incremented modulus under multiple filler may be higher when compared with matrix material [112]. Alternative relevant reason may be incremented percent in stiffness of laminates with different fillers when compared with less structured portion which can considerably improve the modulus share of the polymeric matrices to the laminate modulus [113]. This share of crystallites is also apparent in modulus quantities for EATA filler contents. It can also be examined from Fig. 4 (c) that the reinforcement of multiple fillers also increased the related elongation at break and yield strength value of the laminate when compared with reference sample. The reason assigned to this increment in elongation values emerges from the incorporation of softer inorganic fillers in the epoxy polymeric matrix. The inorganic fillers are flexible than the polymeric resin, hence deform more which leads to decrease the brittleness of composite laminates and finally tends to higher elongation at break [114]. The highest and lowest quantities of elongation at break are formed for L7GEATAER and L7GER samples, respectively, which is similar with the other research findings [14,16,108].

3.6. Effect on tensile fracture characteristics

The microstructural study of epoxy hybrid composites is performed using SEM images to appraise the distribution, and suitability of hydrophilic epoxy matrix with hydrophobic inorganic fillers. The microstructures of the developed polymeric composites such as L7GER,

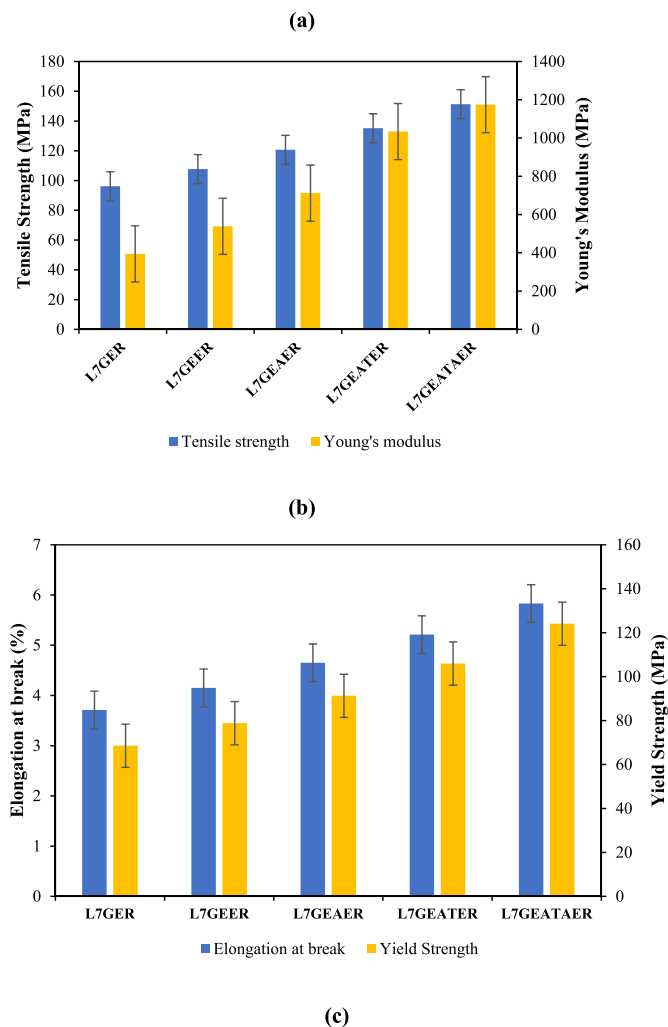
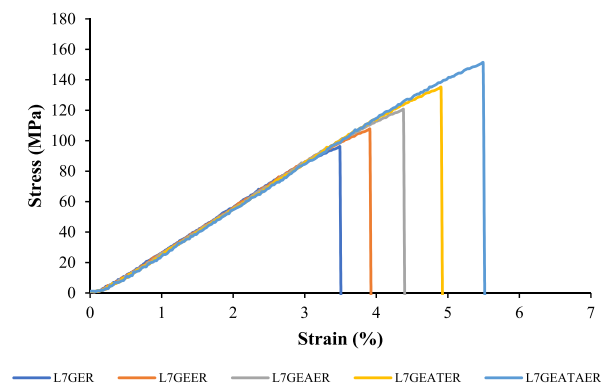


Fig. 4. Tensile properties of $\text{Al}_2\text{O}_3/\text{TiO}_2$ /egg shell/ATH reinforced glass fabric-epoxy hybrid composites (a) Tensile stress-strain, (b) Tensile strength and modulus, (c) Elongation at break and Yield strength.

L7GEER, L7GEAER, L7GEATER, and L7GEATAER are presented in Fig. 5 (a)–(e), respectively. The sample L7GER shows brittle characteristics as there is no reinforcement of fillers within the matrix (Fig. 5 a). The addition of egg shell in the epoxy resin is depicted in Fig. 5 (b), where the egg shell micro-particles are noticed to be isolated and no accumulation are present, hence it has a better distribution within the matrix. However, under egg shell and alumina filler reinforcement, compact scheme of filler particles is observed as presented in Fig. 5 (c). The connection between epoxy matrix and fillers amplifies with the addition of alumina

that tends to improvement in tensile characteristics when compared with L7GER and L7GEER samples. At egg shell, alumina and titania reinforcement, the epoxy adapts the adequate quantity of fillers within the limit as presented in Fig. 5 (d). At this phase, the connection between egg shell, alumina, titania and epoxy resin is optimal, tends to higher tensile strength, modulus, and elongation at break. Further reinforcement of EATA fillers to the epoxy polymer outcome in the tight packing and uniform dispersion of fillers throughout the surface with lower fiber pull outs which prevents the entrapment of moistures from epoxy polymer within the fillers. In this condition, the interaction is good and cavity production is less becomes apparent as examined in Fig. 5 (e). Finally, the result is decreased water absorption and increased tensile characteristics [14,16,115].

3.7. Effect on compression characteristics

Fig. 6 (a) exhibits a stress-strain curves of epoxy hybrid composites under axial compressive load and their respective compressive strength and modulus are depicted in Fig. 6 (b). Reference composite sample presents lower strain, strength and modulus among other composite combinations. Potentially the existence of amorphous region in the without filler could have lower strength and modulus values. The egg shell reinforced laminates exhibits improvement in modulus and strength quantities when compared with reference samples, but the strain value was improved slightly, because of presence of crystalline structure [116–118]. The L7GEAER, L7GEATER, and L7GEATAER composites generated higher compression strength and modulus quantities by 1.29, 1.48, and 1.69 times when compared with reference samples. Whereas the strain value was also increased among all the laminate combinations. The enhanced compression characteristics values may be assigned to increased filler-fabric-matrix interfacial bonding characteristics [19,119]. Since, the egg shell is hard, higher strength scheme, and crystalline which outcome in improved modulus and strength quantities. Additionally to incremented egg shell region, the existence of inorganic fillers such as alumina, titania, and ATH could have also affected further enhanced modulus and strength values in L7GEAER, L7GEATER, and L7GEATAER composites. These inorganic fillers is much harder and higher modulus and strength that could outcome in beyond higher compression characteristics [120].

3.8. Flexural characteristics

The flexural characteristics of epoxy hybrid composites in three point flexural mode are described employing the reinforcement of E, EA, EAT, and EATA fillers. The flexural stress-strain for epoxy hybrid composites along with reference sample is shown in Fig. 7 (a). It is obvious from the curve, the addition of fillers outcome in the steady increment of flexural strength and modulus of epoxy hybrid laminates. These developments can be described on the core of enhanced filler-fabric-matrix interaction with the incorporation of egg shell and inorganic fillers. The more extensive interactions happen to be necessary source for efficient load transfer within matrix to fillers and glass fabrics and strength securing of the laminate [121,122]. It is significant that both flexural and tensile characteristics are steadily incremented with the reinforcement of EATA fillers. It appears that no filler accumulation under multiple filler reinforcement is not as adverse for the characteristics in both axial and lateral loading. Such unique deviation in flexural and tensile characteristics is reported by other researchers [113]. The highest value of flexural strength and modulus for L7GEATAER sample is 624.49 MPa, and 8498.52 MPa, respectively, shown in Fig. 7 (b). The flexural modulus of the developed composites is higher than the corresponding value of other polymer composite, because of the generation of stress concentration location under higher filler density, emanating in early crack growth and eventually in damage [123,124]. The flexural modulus of developed epoxy polymer hybrid composite is observed to be incremented by 1.21–2.14 times when compared with reference

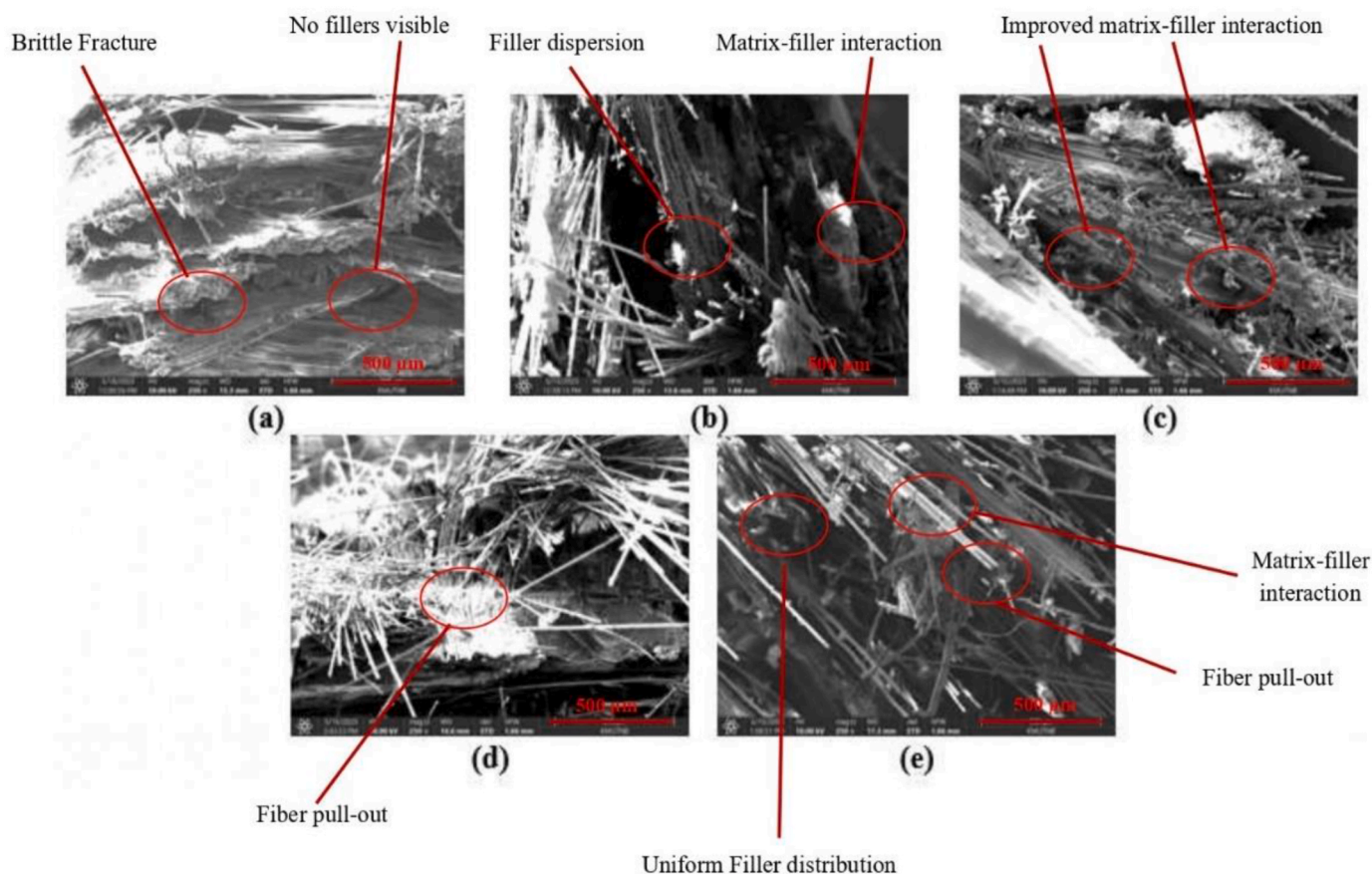


Fig. 5. Tensile fracture of $\text{Al}_2\text{O}_3/\text{TiO}_2$ /egg shell/ATH reinforced glass fabric-epoxy hybrid composites (a) L7GER, (b) L7GEER, (c) L7GEAER, (d) L7GEATER, and (e) L7GEATAER.

samples, hence demonstrating the possible reinforcing result of egg shell and inorganic fillers.

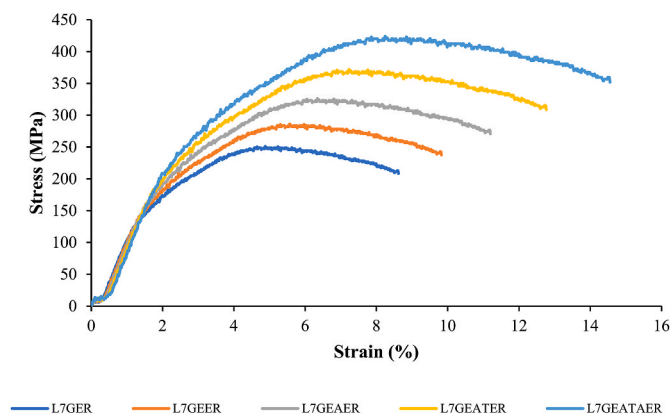
3.9. Interlaminar shear strength

Fig. 8 (a) and (b) exhibits the stress-strain curves and mean interlaminar shear strength of the epoxy hybrid composites with different fillers such as E, EA, EAT, and EATA as evaluated from short beam strength tests. Compared with the reference sample, the mean interlaminar shear strength values improved from 40.07 to 44.47, 49.37, 54.8, and 60.83 MPa with the reinforcement of E, EA, EAT, and EATA fillers, respectively. It is prominent that the interlaminar shear strength of epoxy hybrid composite is primarily evaluated from shear characteristics of the matrix and filler-glass fabric-matrix strong interfacial bonding present the similar filler weight proportion [125,126]. These connections act as an essential roles in shifting load more efficiently within the polymeric matrices and avoiding propagation of crack by deferring growth of micro-crack, amalgamation and hence composite breaks. The incorporation of micro-sized particles to the epoxy polymer could also increment thermal residual loads on the filler surface, and hence the filler-glass fabric-matrix interfacial bonding tends to enhanced interlaminar shear strength [127,128]. Moreover, inorganic fillers such as alumina, titania, and ATH with oxygen functional groups on the monolithic layer microstructure can act as a transfer mode from inorganic-natural fillers to organic polymer contains epoxy higher molecular weight component, linking the distance between matrix and inorganic fillers within the interfacial regions, and hence enhancing the strength to certain range.

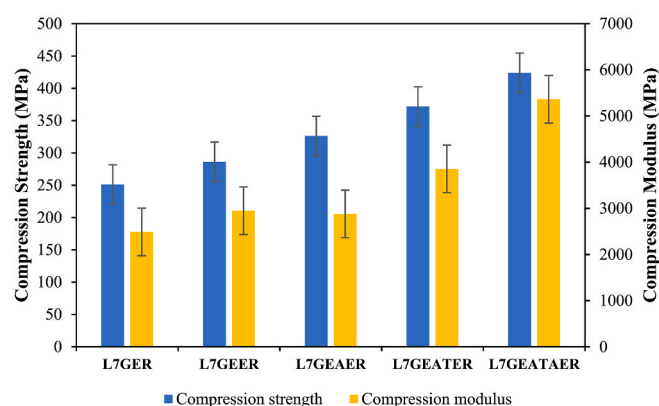
3.10. Impact strength and shore D hardness

The izod impact experiment is most suitable impact conditions which determines the characteristics of a composite subject to dynamic loading like torsion, tension and bending. These tests were broadly used in industries because of the ease of specimen formation and potential to develop comparative information promptly [129–131]. The production of specimen in izod impact experiment is performed as unnotch, where the impact energy is usually much higher than the notched samples, due to in unnotched samples, the evaluated impact strength is influenced from the energy required for crack propagation and initiation. Fig. 9 presents the impact strength of epoxy hybrid composites as a formulation of types of reinforcements. The outcomes exhibit enhancement on the impact characteristics of epoxy composites because of addition of E, EA, EAT, and EATA microparticles in the epoxy polymer. A lower value of impact strength is observed in sample L7GEER when compared with other samples that is assigned to the accumulation of microparticles in laminates. These characteristics can be explained with many parameters. Initially, the magnitude of stress concentration around microparticle must be low, since no decomposing influences on impact strength examined from micro-particles addition in epoxy laminates. Also, it is very essential to select a suitable mixing principle to attain uniform distribution of micro-particles in the epoxy polymer. Reinforcing of micro-particles is another factor which may describe the decomposing influence on impact strength due to it improves the possibility of accumulation in laminates, incrementing the intensity of concentration of stress in composites [129].

The shore D hardness test was performed on epoxy hybrid composites reinforced with E, EA, EAT, and EATA fillers. The average values were collected and presented in Fig. 9. Furthermore, the reinforcement



(a)



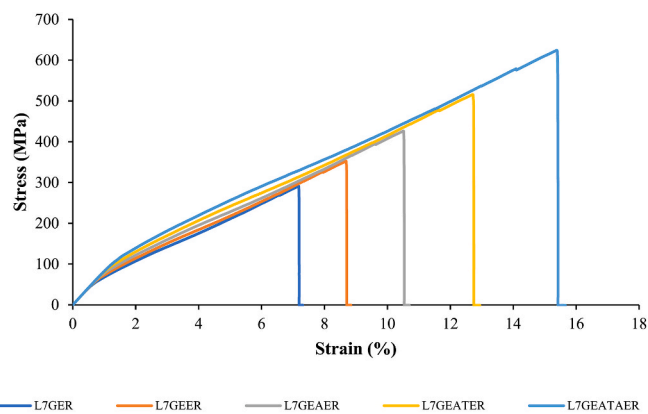
(b)

Fig. 6. Compression properties of $\text{Al}_2\text{O}_3/\text{TiO}_2/\text{egg shell}/\text{ATH}$ reinforced glass fabric-epoxy hybrid composites (a) Compression stress-strain, (b) Compression strength and modulus.

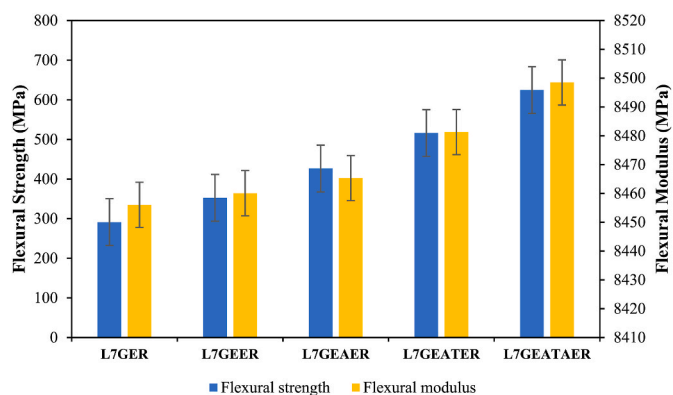
of EATA fillers improved the hardness value by 22.55 % assigned to homogeneous filler dispersion on the sample surface, which tends to substantial hardness deviations, as signified from higher standard difference for every specimen. These findings depicting that more inorganic micro-particles are reinforced in the epoxy matrix, accumulation decreases because of decreased in electrostatic and van der Waals forces between the fillers. This principle describes the higher diversion in hardness characteristics with certain regions consisting non-accumulated microparticle where other distributed filler completely [130,132].

3.11. FTIR

In this investigation, FTIR is applied to determine the presence of functional groups in prepared specimen and examine effect of E, EA, EAT, and EATA filler contents on structural characteristics. The FTIR transmittance spectrum of the epoxy hybrid composites with and without fillers are presented in Fig. 10. There are certain modification in the spectra of L7GEER composites apparent with addition of egg shell particles. The wavenumbers at 999.25, 1154.41, 1495.62, and 1597.98 cm^{-1} are notable peaks for epoxy hybrid composites that present the CH_3 symmetric, sulfone 2, $-\text{CH}_3$ and $-\text{CH}_2$ bending vibrations, respectively [133]. In the existence of EAT, and EATA fillers, transmittance band at 1718.25 and 1742.64 cm^{-1} examined from FTIR spectrum relates to the carbonyl functional groups of maleic anhydride. For all



(a)



(b)

Fig. 7. Flexural properties of $\text{Al}_2\text{O}_3/\text{TiO}_2/\text{egg shell}/\text{ATH}$ reinforced glass fabric-epoxy hybrid composites (a) Flexural stress-strain, (b) Flexural strength and modulus.

specimens, the bands at 816, 995, and 1140 cm^{-1} are examined because of presence of silica structure and glass fabrics which is main reason for improving the mechanical characteristics when compared with reference sample. A wide wavenumber in the portion between 1200 and 825 cm^{-1} is examined for L7GEAER sample with the reinforcement of alumina and egg shell micro-particles. This wavenumber can be ascribed to symmetric stretching of the hydrogen atoms with oxygen molecules improved. From FTIR spectra, it is evident that the addition of inorganic fillers improves the mechanical and physical characteristics of the epoxy hybrid composites [134].

3.12. Thermal stability

Fig. 11 (a) exhibits the weight of the epoxy hybrid composites as a sample temperature function ranging between 30 and 600 $^{\circ}\text{C}$ under nitrogen atmosphere. As it can be observed that during the deterioration of the epoxy hybrid composites in this atmosphere, a rapid decrement initiates around 300–315 $^{\circ}\text{C}$, and this assigned to the primary weight loss of the epoxy polymer. The different trends examined among the TGA experiment are primarily because of the varying compositions of inorganic fillers that modify the thermal decomposition trajectory of the polymer matrices. This multi-stage degradation characteristics signifies different bond-breaking principles emerging at particular temperature ranges for every configuration [135,136]. TG decomposition of the epoxy matrix, as the glass fabrics are not thermally influenced in this temperature, in the subsequent atmosphere participates with a lower

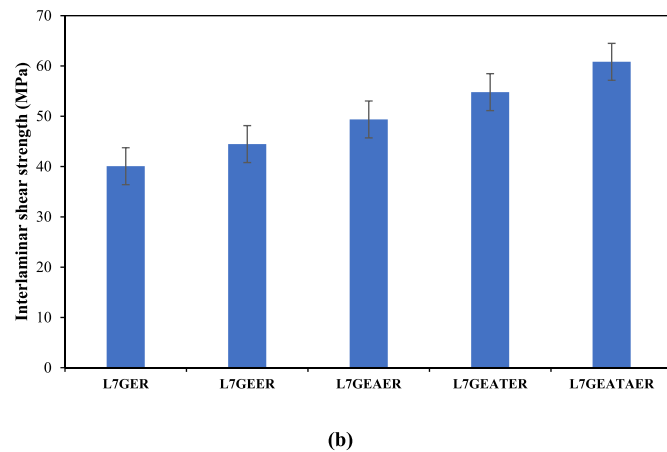
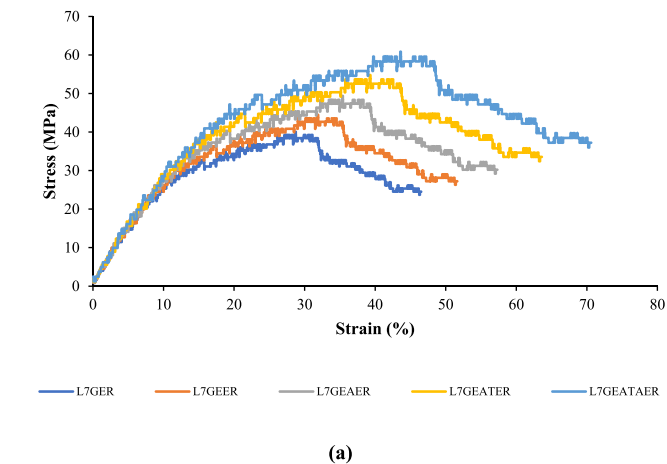


Fig. 8. Interlaminar shear strength of $\text{Al}_2\text{O}_3/\text{TiO}_2/\text{egg shell}/\text{ATH}$ reinforced glass fabric-epoxy hybrid composites (a) stress-strain, and (b) Interlaminar shear strength.

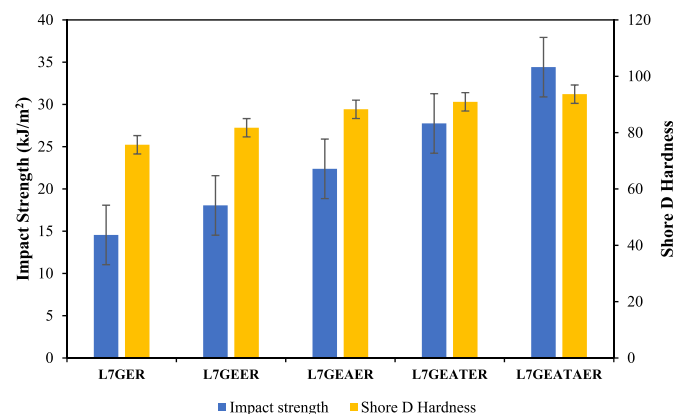


Fig. 9. Impact strength and Shore D hardness of $\text{Al}_2\text{O}_3/\text{TiO}_2/\text{egg shell}/\text{ATH}$ reinforced glass fabric-epoxy hybrid composites.

range and sustains up to final 600 °C. A noticeable point in the TG curves, is the highest drop that relates to the maximum decomposition rate of the examined composites. In the condition of L7GEAER, L7GEATER, and L7GEATAER composites, one peak is observed at 307.2, 327, and 335.2 °C, respectively that is assigned to onset temperature and decomposition of the epoxy matrix bond which is comparably higher than the reference sample [137–139]. The five repetition of TGA tests were performed to assure statistical accuracy and affirm the

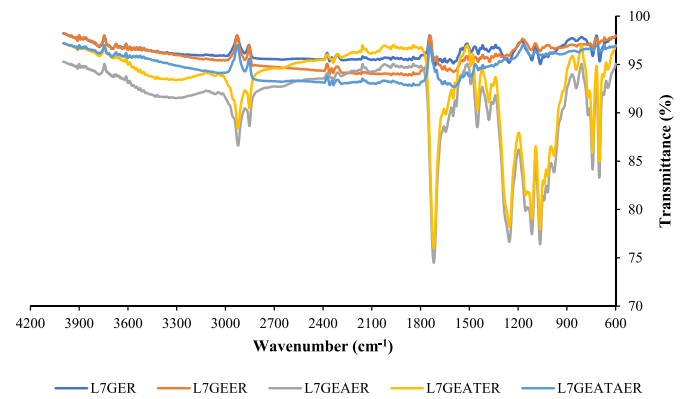


Fig. 10. FTIR spectrum of $\text{Al}_2\text{O}_3/\text{TiO}_2/\text{egg shell}/\text{ATH}$ reinforced glass fabric-epoxy hybrid composites.

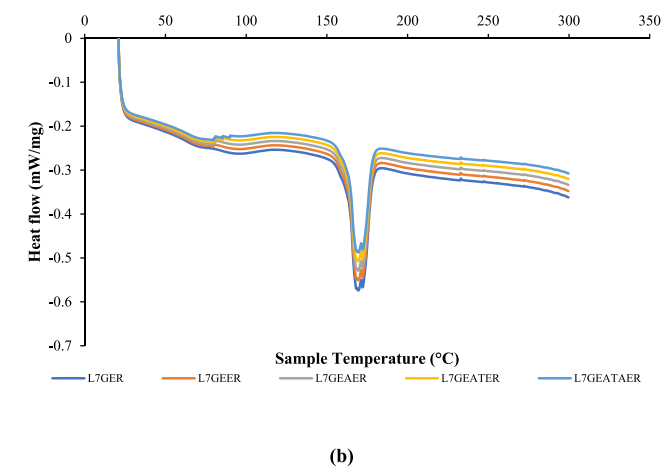
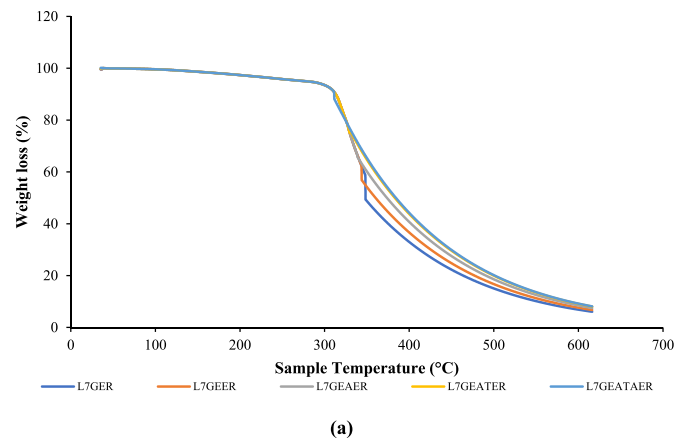


Fig. 11. Thermal stability characteristics of $\text{Al}_2\text{O}_3/\text{TiO}_2/\text{egg shell}/\text{ATH}$ reinforced glass fabric-epoxy hybrid composites (a) TGA and (b) DSC.

reproducibility of thermal characteristics across different configuration. These repeated experiments showed that the addition of multiple inorganic particles produced a synergistic influence, arising in the staggered degradation profiles [140,141]. It was observed in the recent literature that the addition of nano fillers in epoxy polymer affect the degradation temperature of the hybrid composites, since alumina, titania and ATH exhibits better thermal characteristics [142]. It can also be observed that the specimen consisting E, EA, EAT, and EATA improved the endset temperature of the developed composites by nearly 4.8, 5.7, 27.1, and 59.3 °C, respectively, when compared with reference sample. This increment is assigned to the incorporation of alumina, titania, and ATH

in combined manner which avoids the movement of the polymeric network during the degradation technique [143–145]. The divergent trends in the TGA graphs between 350 and 500 °C specifically highlight the production of char residue with different thermal stabilities, where more rigidi configuration with multiple particles illustrate improved barrier effects which obstruct the diffusion of volatile decomposition constituents [146,147]. The degradation rate of the epoxy hybrid composite present slightly higher when compared with reference sample which signifies the existence of inorganic fillers in the epoxy structure plays as a resistance from flame. After thermal stability examination signifies the addition of inorganic fillers improves the carbon residue [143,148].

The heat flow of the epoxy hybrid composites as a function of sample temperature and different combinations of epoxy hybrid composites are shown in Fig. 11 (b) and Table 3. The different DSC profiles collected from 5 repeated testing examine different variations in the width and depth of the endothermic peaks, indicating deviations in heat capacity and crystallization characteristics of the various configurations. These repetitions were necessary to validate that the examined deviations were ascribable to material characteristics rather than experimental differences [149–151]. The quantity examined for the glass transition temperature (T_g) of the reference sample (L7GER) was 161.25 °C, whereas the T_g of samples such as L7GEER, L7GEAER, L7GEATER, and L7GEA-TAER improved by 1.99, 4.03, 7.16, and 8.92 %, respectively. The reason may be assigned to the increment of the crosslinking and proper order of matrix cure reaction because of the existence of the inorganic fillers [152], which is homogeneously dispersed within the matrix. The different trends examined in the DSC graphs, specifically between 150 and 250 °C, can be assigned to varying degrees of interfacial interactions within the epoxy matrices and the different combinations of fillers that produce identical energy absorption principles during the transition of phase [153,154]. Another reason is the improvement in cross-link density, mechanical incorporation emerging from a strained inorganic fillers microstructure and limitation of sectional relaxation of network division around the inorganic fillers [24,96,111,155]. The binding density is correlated with the distribution and quantity of micro-particles in the epoxy laminates [156]. The considerable deviations in heat flow profiles after the main transition around 150 °C illustrate the production of different morphological scheme with different thermal resistance, where multiple-filler system produce more rigid network framework which require higher energy for molecular movement [157–159]. The movement of epoxy network near the inorganic fillers are limited which to improvement in thermal stability [160]. Hence, the fabricated epoxy hybrid composites can be applied under maximum temperature than the reference samples. Any modification in mechanical characteristics are because of the physical existence of inorganic fillers and also to modify in the polymeric chain [161]. The repetition of five DSC tests affirmed that the shifts in baseline heat flow between −0.2 and −0.5 mW/mg across different configurations relate to systematic modification in heat capability ascribed with incremental alteration to the polymer structure, verifying the progressive improvement in thermal characteristics [162,163]. Hence, the addition of multiple fillers tends to higher glass transition temperature of the developed epoxy hybrid composites investigated.

Table 3

Thermal stability of Al₂O₃/TiO₂/egg shell/ATH reinforced glass fabric-epoxy hybrid composite.

	Glass transition temperature	Onset temperature	Endset temperature	Char residue
L7GER	161.25	292.6	352.6	6.1
L7GEER	164.47	305	357.4	6.7
L7GEAER	167.76	307.2	358.3	7.5
L7GEATER	172.79	327	379.7	7.9
L7GEATAER	175.63	335.2	411.9	8.3

3.13. Predictive analysis

3.13.1. Model performance and dynamics of learning

The evaluation of model performance starts with an analysis of validation and training loss graphs over training epochs as illustrated in Fig. 12. These graphs provide observation in the dynamics of learning deep neural network techniques. Both plots exhibit a persistent recession trend with incrementing epochs, signifying efficient learning and separation of scheme from the datasets. Lower final loss quantities for both validation and training sets around 0.018 and 0.015 recommends higher predictive precision throughout the epoxy hybrid composite characteristics. Intensive monitoring of training and validation loss plots, specifically in upcoming epochs, signifies better generalization without overfitting. Smooth drop of both plots implies a steady learning technique, assigned to suitable rate of learning and efficient regularization methods [164,165].

Mean absolute error assessment for validation and training sets enhances the loss plots as shown in Fig. 13. These plots for both sets reduce persistently, imitating the loss graph trends. Lower final mean absolute error quantities for validation and training sets around 0.09, and 0.08 respectively, signify mean forecasting variation of lower than 10 % of the characteristic range. The compact gap between validation and training mean absolute error (around 0.01) affirms better generalization to invisible data. The consistency between mean absolute error loss and trend graphs confirms model performance throughout distinct assessment indicators.

R-squared quantities for every forecasted characteristic are examined as given in Fig. 14. R-squared quantities range between 0.77 and 0.98, showcasing the multi-faceted issues in forecasting different material characteristics. All characteristics exhibit remarkably higher R-squared values (>0.96), which exhibit better overall forecasting performance throughout different epoxy hybrid composites. Error evaluation is critical for understanding a limitations and strength of models in forecasting different epoxy hybrid composite characteristics. Two contributing error indicators are considered such as root mean squared error and mean absolute error. A bar chart of mean absolute error shown in Fig. 15 for every predicted characteristic offers insights of accuracy of predicted models. These values range between 0.017 and 11.75, with most characteristics exhibiting mean absolute error below 6.0 signifying better forecasting precision. Best forecasted characteristics contain mean absolute values of density, porosity, glass transition, onset, and endset temperatures as 0.017 kg/m³, 1.5 %, 1.3 °C, 1.8 °C, 0.97 °C, respectively. Lower mean absolute error values signify prediction nearer to original quantities on mean. The deviation in mean absolute error throughout the characteristics recommends certain epoxy hybrid composite characteristics are intrinsically tougher to predict than others. The root mean squared error (Fig. 16) chart exhibits a similar scheme to the mean absolute error graph with values ranging between 0.022 and 17.5. For most characteristics, root mean squared error is slightly higher than mean absolute error, with this deviation more noticeable for characteristics such as flexural and tensile strength. The consolidated examination offers a sophisticated understanding of accuracy of predictive model. The lower mean absolute and root mean squared error signifying better overall prediction precision. Mechanical characteristics such as flexural and tensile strength exhibit higher mean absolute and root mean squared error potentially because of intricate parameter interplay or broader quantity extents. Characteristics with lower mean absolute error but moderately higher root mean squared error (young's modulus, compression modulus, and flexural modulus) recommend usually precise predictions with intermittent higher variations [166–168]. Characteristics such as density, porosity, glass transition, onset and endset temperatures exhibit persistently lower error throughout both indicators, signifying high-accurate forecasting. This comprehensive error evaluation offers insights in the predictive models capacities throughout distinct epoxy hybrid composite characteristics, substantiating strong overall performance and determining particular

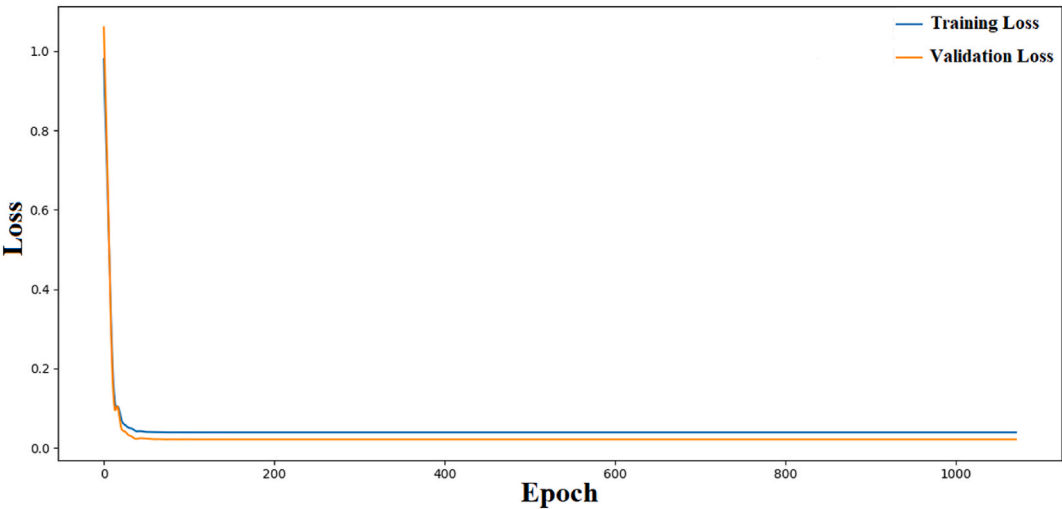


Fig. 12. Model performance curve between training and validation loss.

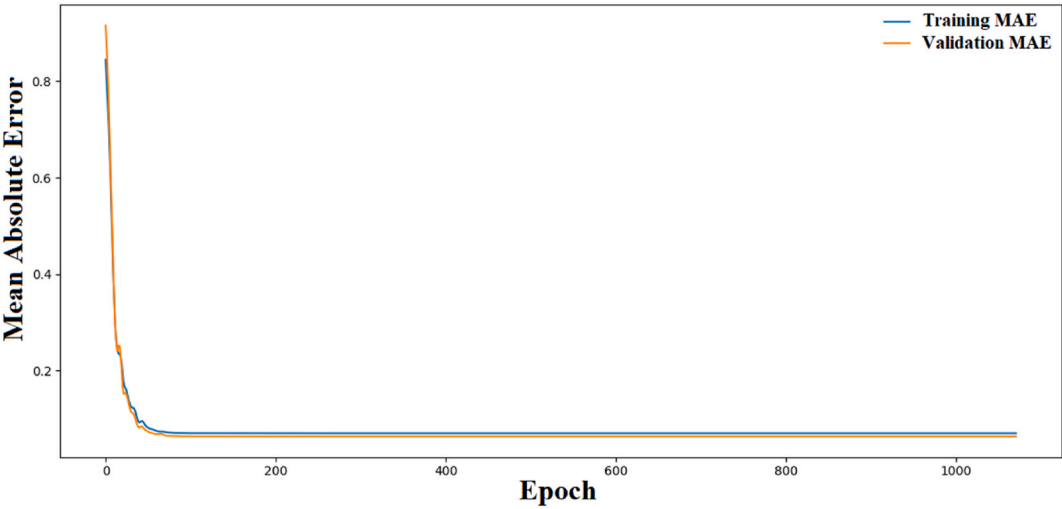


Fig. 13. Mean absolute error performance curve between training and validation loss.

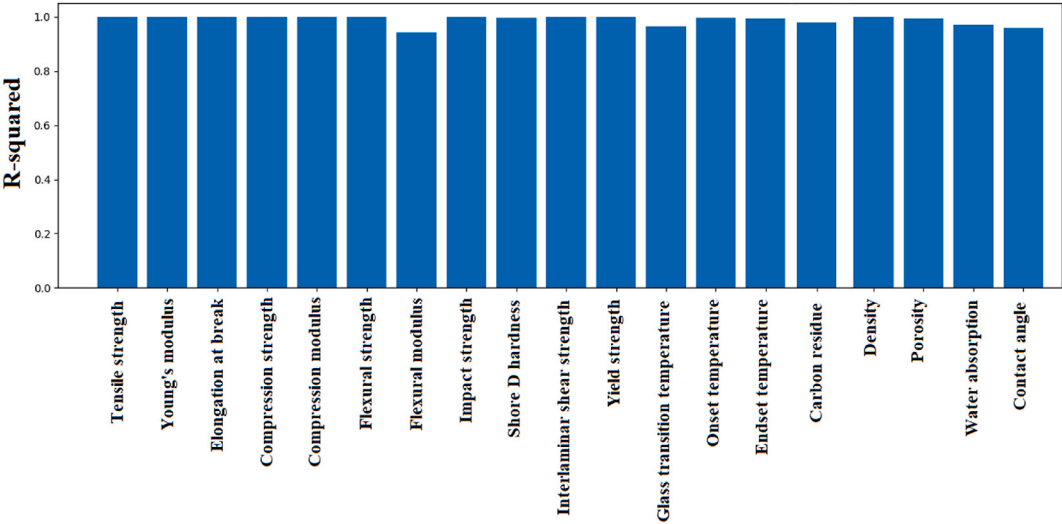


Fig. 14. R-squared value of different properties of Al₂O₃/TiO₂/egg shell/ATH reinforced glass fabric-epoxy hybrid composites.

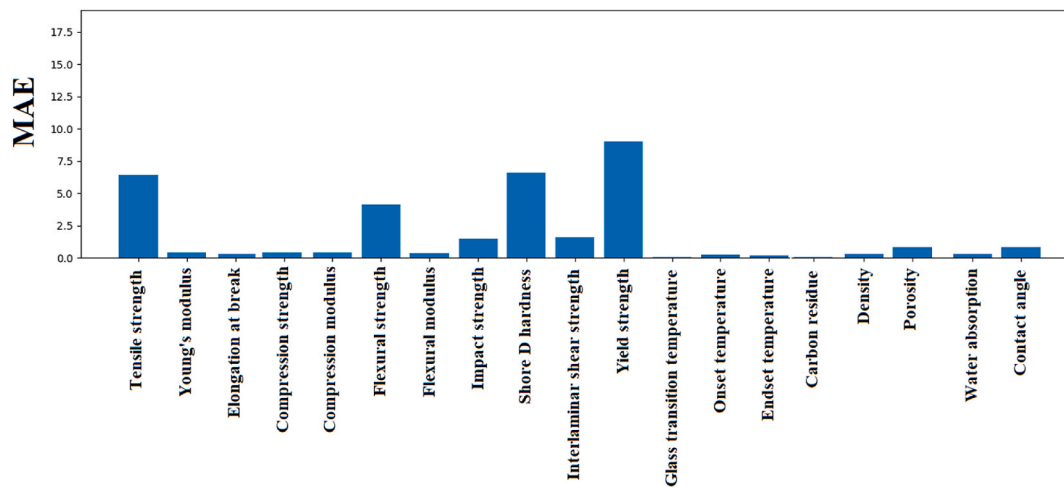


Fig. 15. Mean absolute error of different properties of $\text{Al}_2\text{O}_3/\text{TiO}_2/\text{egg shell}/\text{ATH}$ reinforced glass fabric-epoxy hybrid composites.

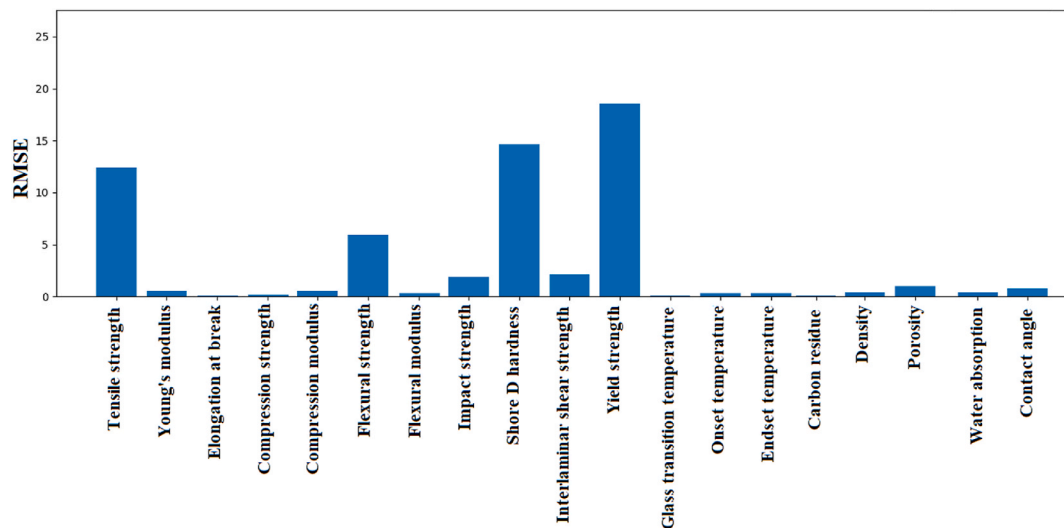


Fig. 16. Root mean squared error of different properties of $\text{Al}_2\text{O}_3/\text{TiO}_2/\text{egg shell}/\text{ATH}$ reinforced glass fabric-epoxy hybrid composites.

regions where prediction precision fluctuates.

3.13.2. Correlation analysis

The investigation of epoxy hybrid composites deploys an improved heatmap illustration to examine correlations between different characteristics is shown in Fig. 17. This assessment tool offers perspectives into the connections among distinct properties of epoxy hybrid composites, providing support for composite optimization and design procedures. The coefficients of Pearson relationship are estimated within all sets of characteristics in the dataset. These coefficients varying between -1 and 1 , signify the direction and strength of linear correlations within characteristics. The emerging correlation matrix is depicted as a heatmap with colour intensity portraying strength of relationships and tone signifying negative or positive relationships. Many characteristics present strong positive correlations ($r > 0.8$), observed among mechanical characteristics such tensile strength, young's modulus, elongation at break, compression strength, flexural strength, impact strength, shore D hardness, interlaminar shear strength, and yield strength which is indicated with dark blue squares in the heatmap. This collection recommends that enhancements in one of these characteristics are likely to be assisted by improvements in others. A noticeable positive relationship is observed within glass transition temperature and density, which coordinates with basic mechanisms of composites, where free electrons

frequently share to both physical and thermal characteristics. Many dark red squares in the heatmap signify strong negative correlations, which are observed within density and particular characteristics such as strength and modulus. This reverse link is predicted, as particular characteristics are estimated from separating the absolute characteristic by density. A moderate to strong negative correlation is examined between the glass transition temperature and compression modulus. This recommends a trade-off within the strength and thermal stability in design of epoxy hybrid composites [169–171]. Several characteristics exhibit weak or no correlations, showcased from light-colored squares in the heatmap, which recommends that flexural modulus and endset temperature can possibly be tuned without considerably affecting other mechanical characteristics. The glass transition and onset temperature shows weak correlations with other characteristics, signifying that it may be affected from parameters not highly associated to other evaluated properties.

3.13.3. Regression analysis

Regression curves for every characteristic offer an image illustration (Fig. 18a–s) of the performance of predictive model, providing perceptions in the precision and stability of prediction throughout the different composite properties. Many characteristics explain better predictive performance, identified by compact clustering near the diagonal line.

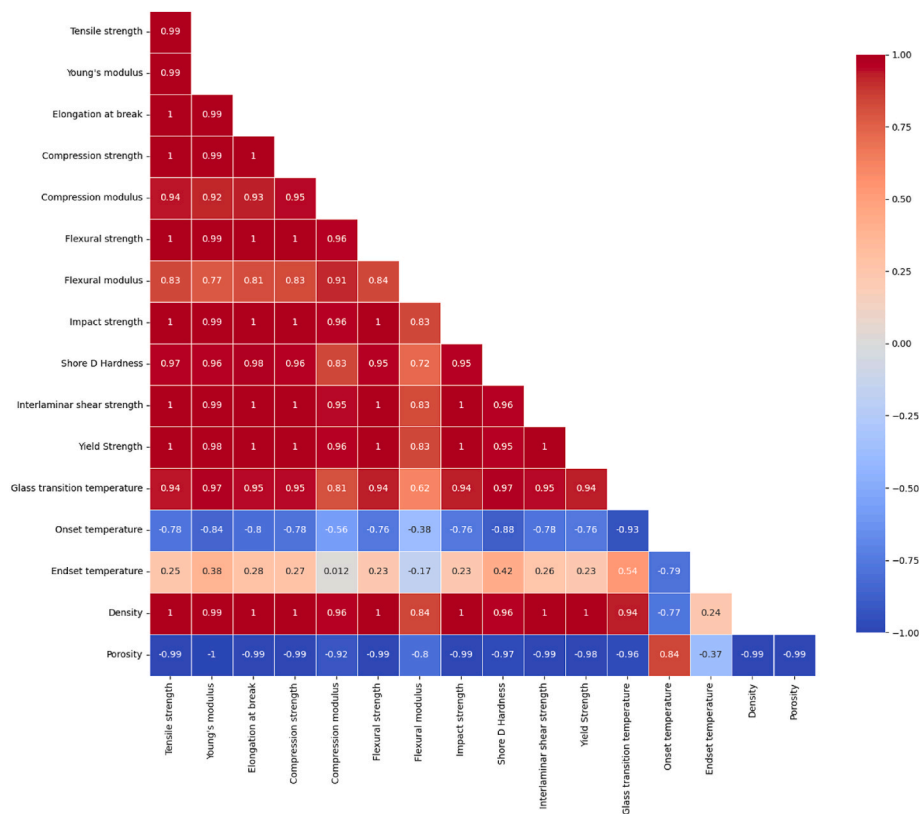


Fig. 17. Correlation heat map of different properties of $\text{Al}_2\text{O}_3/\text{TiO}_2/\text{egg shell}/\text{ATH}$ reinforced glass fabric-epoxy hybrid composites.

The young's modulus exhibits very comfort gathering along the diagonal with values immediately trailing the unique prediction line which coordinates with higher R-squared value as 0.976 and signifies higher precision predictions throughout the complete range of quantities. Tensile strength shows compact segregating with slight scatter affirming its higher R-squared as 0.962 and illustrating the strong capacity of models in forecasting this crucial mechanical characteristic. Flexural strength presents precise synchronization, assisting the higher R-squared value as 0.958 and signifying reliable prediction precision. Glass transition temperature reveals better alignment with the diagonal but with prominent scatter, specifically under higher values. This signifies usually better prediction with scope for enhancement in grasping extreme quantities. Density forecasting are highly precise throughout the complete range as indicated from the compact segregation in the regression curve and lower root mean squared and mean absolute error values. The void concentration forecasting, while generally usual, exhibit incremented scatter, signifying obstacles in collecting all parameters affecting void production in epoxy hybrid laminates.

3.13.4. Best composite selection

The epoxy hybrid composite design and selection commonly includes stabilizing multiple, sometimes contradictory, material characteristics. An intricate MCDm method is established to handle this problem by applying radar graphs and overall score ratings for comprehensive examination of epoxy hybrid composites. Different epoxy hybrid composites balance different characteristics distinctly. Symmetrical polygons signify better balance throughout all characteristics, where asymmetrical shapes recommend strength in particular regions and weaknesses in others which is shown in Radar plot of Fig. 19. The highest-rated material outperforms in diverse characteristics, specifically mechanical properties. The graph depicts trade-off between characteristics, such as the reverse relationship between porosity and water absorption [172]. The bar chat shown in Fig. 20 determines the best-performing materials with the highest-rated composite attaining an

overall score of 0.892 and observed that the sample L7GEATAER is best among the others. A slight reduction in scores throughout the composites recommends few competitive effectiveness. Considerable gaps within some composites score would signify natural critical points in performance sections. The ranking possess determination of reference composite as L7GER by defining standards in the dataset. Thus, the combination of overall scores and radar plots offers a comprehensive toolkit for selection of best composite, providing supplementary observations, verifying the scoring techniques, and enabling profound insight of composites with similar cumulative scores [173–176].

3.14. CNN analysis

The CNN-based assessment of SEM micrographs has generated an extensive set of quantitative outcomes across five epoxy hybrid composite samples, offering deep perception into their morphological properties. This detailed analysis exhibit complex deviations in multiple factors, providing a sophisticated understanding of the potential performance and composite characteristics. Average intensity and their standard deviation exhibit considerable variation throughout the epoxy hybrid composite samples. The sample L7GEATAER illustrates the maximum intensity (Fig. 21a and b) of 0.56 with standard deviation of 0.30, whereas the specimen L7GEATER exhibits lower intensity of 0.32 and sample L7GER with lower standard deviation of 0.21. These deviations recommends divergences in composition homogeneity and interaction of electron among the samples. The maximum intensity in L7GEATAER may signify a higher existence of inorganic fillers with incremented surface conductivity and higher atomic numbers. The roughness estimates (Fig. 21 c) shows lower deviation when compared with intensity estimates, ranging between 0.044 and 0.028 with L7GEATAER sample exhibiting the maximum roughness. The fracture estimate presents (Fig. 21 d) a reducing trend from L7GER to L7GEATER (0.126–0.087) before incrementing again for L7GEATAER sample. These factors provide quantitative evaluation of possible mechanical

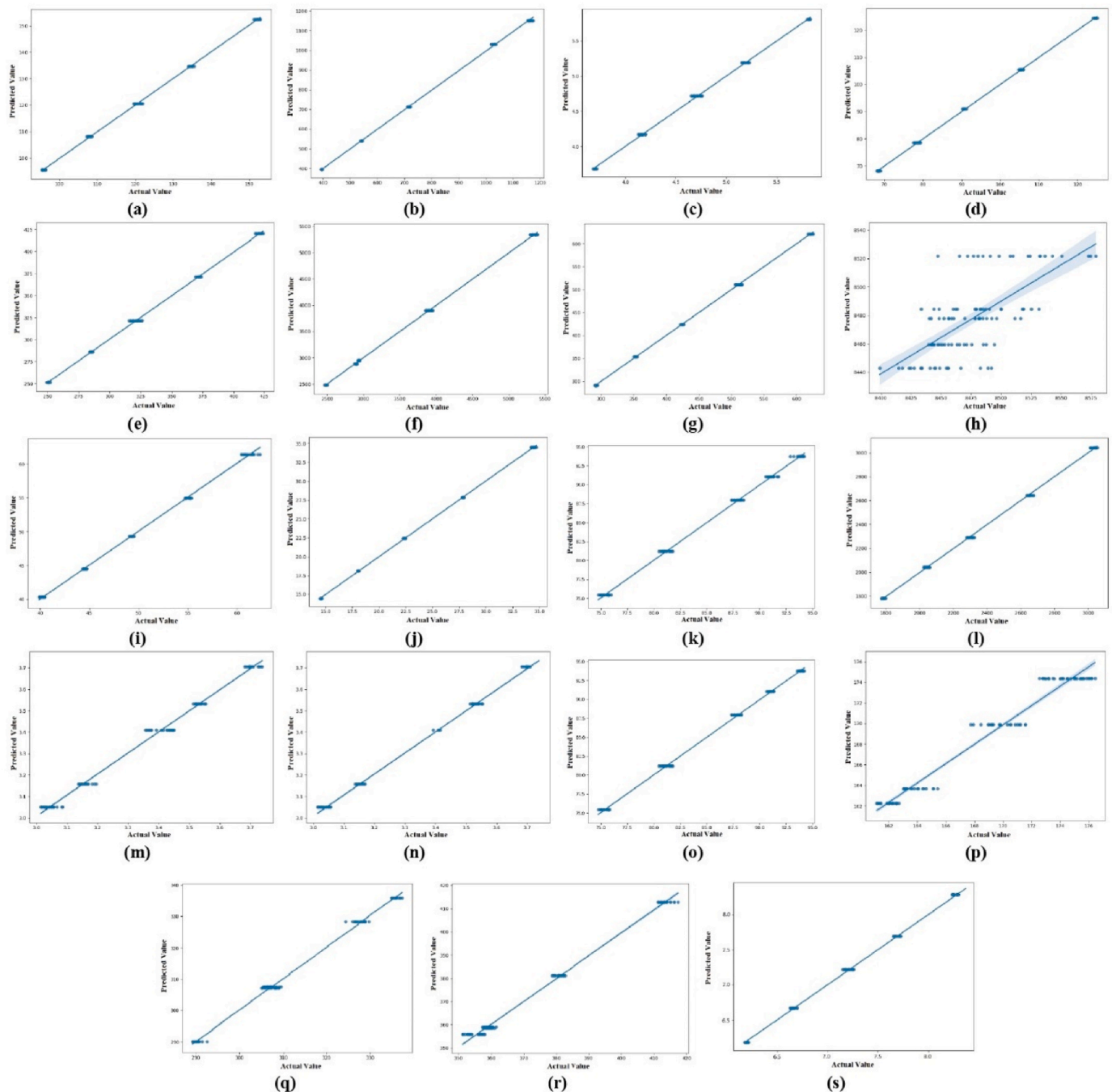


Fig. 18. Regression analysis of epoxy hybrid composites (a) Tensile strength, (b) Young's modulus, (c) Elongation at break, (d) Yield strength, (e) Compression strength, (f) Compression modulus, (g) Flexural strength, (h) Flexural modulus, (i) Interlaminar shear strength (j) Impact strength, (k) Shore D hardness, (l) Density, (m) Porosity, (n) Water absorption, (o) Contact angle, (p) Glass transition temperature, (q) Onset temperature, (r) Endset temperature, (s) Carbon residue.

characteristics and surface topography. The reverse relationship between fracture and roughness estimates in L7GEATER and L7GEATAER samples recommends a potential trade-off between fracture resistance and smoothness of surface. GLCM factors such as homogeneity, correlation, energy and contrast explain rigid textural deviation through the samples are shown in Fig. 21(e)–(h), respectively. Remarkably, GLCM energy exhibits an incrementing trend from sample L7GEAER, L7GEATER, and L7GEATAER, recommending a development towards repetitive or organized textures in these specimens. These deviations signify divergences in uniformity and texture complexity among the specimens. Morphological characteristics present significant differences. The sample L7GEER has highest objects (perimeter and area as 36

μm and $760 \mu\text{m}^2$) (Fig. 21i and j), whereas the sample L7GEATAER has smallest (perimeter and area as $25 \mu\text{m}$ and $137 \mu\text{m}^2$). These deviations in grain or particle sizes could possess essential effects for epoxy hybrid composite characteristics such as strength, modulus, and thermal stability. The sample L7GEER may depict a composite with coarser grains or accumulated fillers, whereas the sample L7GEATAER could signify a finer microstructure. The FFT energy (Fig. 21 k) outcomes reveals significant deviations, highest at L7GEER (6.9×10^5) and attaining its minimal value at L7GEATER (2.7×10^5). This differences, specifically the higher for L7GEER may signify the existence of long-range order or periodic scheme, potentially correlated with regular surface schemes, and crystalline structures. The statistical analysis offer additional

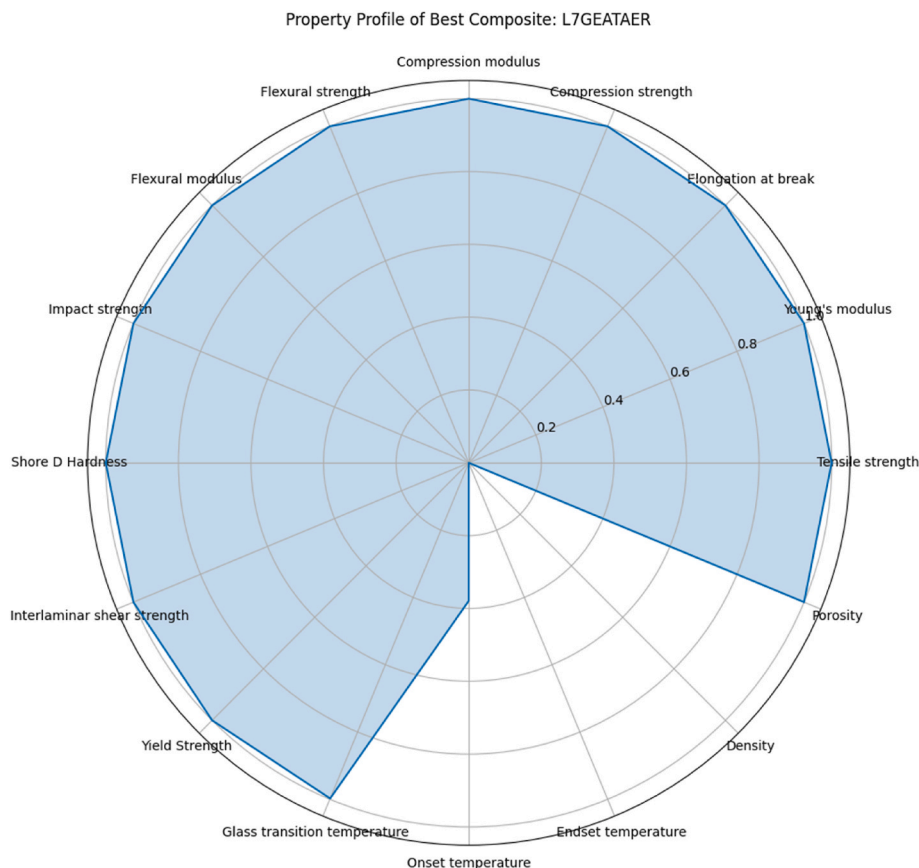


Fig. 19. Radar plot of best epoxy hybrid composite.

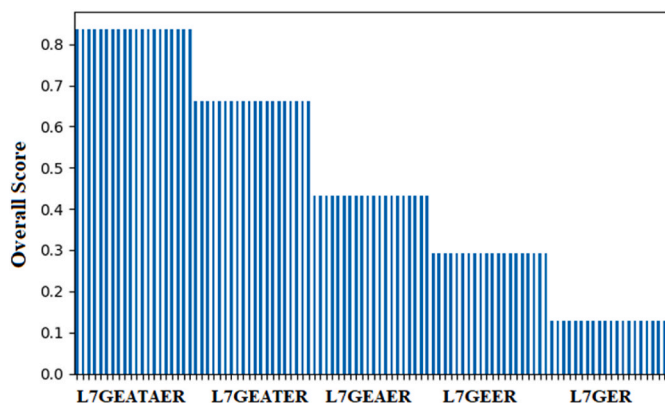


Fig. 20. Overall scores of $\text{Al}_2\text{O}_3/\text{TiO}_2/\text{Egg Shell}/\text{ATH}$ reinforced glass fabric-epoxy hybrid composites.

observations on intensity dispersion. Skewness (Fig. 21 l) varies between -0.08 and 0.80 with mostly positive ranges recommending asymmetric intensity dispersion with extremities expanding towards higher quantities. The reason may be assigned to the presence of inorganic fillers or bright characteristics. Kurtosis (Fig. 21 m) ranges from 1.79 to 2.92 , signifying platykurtic dispersion and recommending a partial deficiency of outliers when compared with normal distribution. Entropy (Fig. 21 n) exhibits lower variations, varying between 6.65 and 7.63 , offering insights in the overall intricacy of the intensity dispersions. The signal-to-noise ratio (SNR) (Fig. 21 o) presents an incrementing trend, decrementing from sample L7GER to L7GEATER before incrementing for L7GEATAER sample. This scheme could be correlated with deviation in sample preparation quality, inherent material characteristics, and

imaging criteria influencing interactions of electron. These insights are critical for evaluating the effectiveness of other collected properties and found that L7GEATAER is best among the other laminates (Fig. 21 p). The CNN-based assessment of SEM micrographs has offered comprehensive quantitative information on sample micrographs, exhibiting complex relationships between processing, composition, and final structure [177–179]. These observations provide valuable direction for exploring structure-property interactions and selecting the processing and material design for specific applications.

3.15. Statistical analysis

The outcomes of physical, mechanical and thermal characteristics were examined statistically from one-way ANOVA method. To identify the influence of inorganic fillers on density, porosity, water absorption, contact angle, tensile strength, young's modulus, compression strength, compression modulus, flexural strength, flexural modulus, impact strength, shore D hardness, glass transition, onset, endset temperature, and carbon residue characteristics of epoxy hybrid composites is performed from ANOVA technique. Table 4 exhibits the ANOVA outcomes for physical, mechanical, and thermal characteristics in terms of between groups and within groups, where significance levels are less than 0.05 . It states that the influence of inorganic fillers on physical, mechanical, and thermal characteristics is statistically significant under the confidence level of 95% [1,131,175,180]. The exceptionally lower p-values illustrate that the deviation among composite scheme are highly considerable and not ascribable to experimental deviation. This rigorous statistical verification across all measured characteristics offers a solid foundation for the subsequent multi-criteria optimization and CNN evaluation, affirming that the superior characteristic of L7GEATAER is statistically reliable.

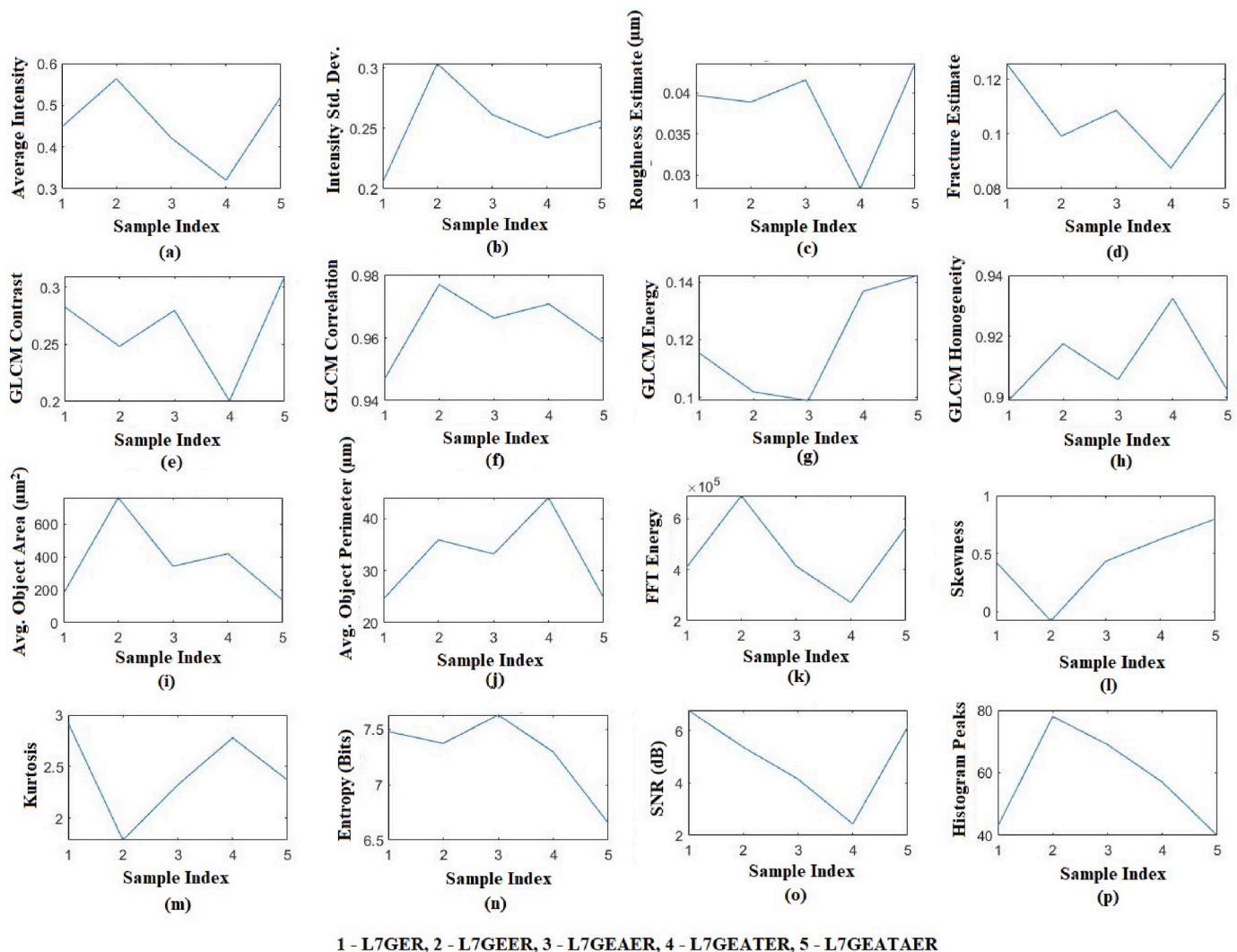


Fig. 21. SEM image analysis of $\text{Al}_2\text{O}_3/\text{TiO}_2/\text{Egg Shell}/\text{ATH}$ reinforced glass fabric-epoxy hybrid composites (a) Average intensity, (b) Intensity standard deviation, (c) Roughness estimate, (d) Fracture estimate, (e) GLCM contrast, (f) GLCM correlation, (g) GLCM energy, (h) GLCM homogeneity, (i) Average object area, (j) Average object perimeter, (k) FFT energy, (l) Skewness, (m) Kurtosis, (n) Entropy, (o) Signal to Noise Ratio (SNR), and (p) Histogram peaks.

5. Possible applications with industrial limitations

These multi-filler hybrid polymer composite exhibit assurance for automotive applications consisting semi-structural, interior panels, and under-hood components where balanced thermal, weight and mechanical characteristics are desired. The improved mechanical performance and thermal stability make them adequate for construction components such as partition boards, ceiling panels, and decorative materials with enhanced fire resistance. In electrical applications, these materials could serve as enclosures and housings requiring better thermal stability and insulation characteristics. Furthermore, industrial implementation faces issues including attaining cost-competitiveness, meeting field-specific standards, and consistency in large-scale production against the well-known materials. The fabrication complexity exhibits additional limitation such as modification of process, and high-viscosity of multi-filler polymer may need specialized device. The quality control remains crucial for assuring consistent filler distribution throughout the batches of production where material supply chain consideration contain establishing efficient sourcing of consistent-quality waste materials. Economic viability must consider for higher initial setup costs against long-term benefits, possibly requiring policy motivation to support waste material valorization. Also addressing, technical knowledge gaps between industry workforce training and generating standardized

processing protocols which is important for extensive selection of these sustainable materials in industrial applications [184–186].

6. Conclusion

This investigation exhibited that adding eggshell waste with inorganic fillers such as Al_2O_3 , TiO_2 and ATH in glass-fabric reinforced epoxy hybrid composites significantly improves physical, mechanical and thermal characteristics. The key findings include:

1. Reinforcement of multiple fillers incremented density by up to 48.2 % while decreasing water absorption and porosity by 25.4 % and 18.8 %, respectively, producing more compact composites.
2. Mechanical characteristics displayed remarkable enhancements with L7GEATAER composite presenting the best overall performance: 151.31 MPa of tensile strength, 423.92 MPa of compression strength, 624.49 MPa of flexural strength, 34.41 kJ/m^2 of impact strength, and 93.65 of Shore D hardness.
3. Thermal stability enhancements were verified by improvement in glass transition, onset temperature and carbon residue by 8.92 %, 12.71 %, and 26.51 %, respectively in the optimal L7GEATAER composite.

Table 4

Statistical analysis of Al₂O₃/TiO₂/egg shell/ATH reinforced glass fabric-epoxy hybrid composites.

		Sum of Squares	Degree of freedom	Mean Square	F-value	P-value
Density (kg/m ³)	Between groups	19528316.794	4	4882079.198	6390.527	1.54×10^{-114}
	Within groups	72575.786	95	763.956		
	Total	19600892.580	99			
Contact angle (Deg.)	Between groups	2335.531	4	583.883	1425.703	7.34×10^{-84}
	Within groups	38.906	95	0.410		
	Total	2374.437	99			
Porosity (%)	Between groups	5.853	4	1.463	2001.376	9.30×10^{-91}
	Within groups	0.069	95	0.001		
	Total	5.923	99			
Water absorption (%)	Between groups	15.293	4	3.823	2958.432	9.68×10^{-99}
	Within groups	0.123	95	0.001		
	Total	15.416	99			
Tensile strength (MPa)	Between groups	38924.495	4	9731.124	5816.892	1.32×10^{-112}
	Within groups	158.926	95	1.673		
	Total	39083.421	99			
Young's modulus (MPa)	Between groups	8280350.373	4	2070087.593	9418.870	1.63×10^{-122}
	Within groups	20879.184	95	219.781		
	Total	8301229.557	99			
Elongation at break (%)	Between groups	55.711	4	13.928	5377.774	5.41×10^{-111}
	Within groups	0.246	95	0.003		
	Total	55.957	99			
Compression strength (MPa)	Between groups	368652.587	4	92163.147	5775.312	1.86×10^{-112}
	Within groups	1516.022	95	15.958		
	Total	370168.608	99			
Compression modulus (MPa)	Between groups	105622145.937	4	26405536.484	10118.086	5.47×10^{-124}
	Within groups	247924.941	95	2609.736		
	Total	105870070.878	99			
Flexural strength (MPa)	Between groups	1353574.145	4	338393.536	8141.695	1.62×10^{-119}
	Within groups	3948.488	95	41.563		
	Total	1357522.633	99			
Flexural modulus (MPa)	Between groups	57922.154	4	14480.538	20.540	3.21×10^{-12}
	Within groups	66974.942	95	704.999		
	Total	124897.096	99			
Impact strength (kJ/m ²)	Between groups	4974.475	4	1243.619	9786.784	2.65×10^{-123}
	Within groups	12.072	95	0.127		
	Total	4986.547	99			
Shore D hardness	Between groups	4465.886	4	1116.471	2171.137	2.03×10^{-92}
	Within groups	48.852	95	0.514		
	Total	4514.738	99			
Interlaminar shear strength (MPa)	Between groups	5509.911	4	1377.478	5165.910	3.62×10^{-110}
	Within groups	25.332	95	0.267		
	Total	5535.243	99			
Yield strength (MPa)	Between groups	39241.850	4	9810.463	8326.678	5.59×10^{-120}
	Within groups	111.929	95	1.178		
	Total	39353.779	99			
Glass transition temperature (°C)	Between groups	2797.606	4	699.402	667.043	1.40×10^{-68}
	Within groups	99.609	95	1.049		
	Total	2897.215	99			
Onset temperature (°C)	Between groups	26227.476	4	6556.869	3602.612	8.96×10^{-103}
	Within groups	172.903	95	1.820		
	Total	26400.379	99			
Endset temperature (°C)	Between groups	48064.117	4	12016.029	4972.513	2.20×10^{-109}
	Within groups	229.567	95	2.416		
	Total	48293.684	99			
Carbon residue (%)	Between groups	65.678	4	16.419	3159.072	4.40×10^{-100}
	Within groups	0.494	95	0.005		
	Total	66.172	99			

- CNN evaluation of SEM microstructure exhibited that L7GEATAER sample exhibited optimized microstructural properties, containing intensity of 0.56 and higher surface roughness of 0.044, with smaller filler dimensions recommending enhanced filler dispersion.
- Deep learning predictive models attained higher accuracy (R-squared >0.96) for most characteristics, depicting a configuration for future composite property prediction with minimal experimental results.

This investigation illustrates the viability of comprising biowaste and inorganic fillers to produce sustainable and high-performance composite materials adequate for challenging applications where consistent thermal, environmental and mechanical characteristics are crucial. From these promising outcomes, industrial implementation faces issues such

as quality control for consistent particle dispersion, cost-competitiveness, processing rigidity because of high-viscosity multi-particle schemes, and the necessity for standardized protocols. Future work should focus on reinforcing additional bio-waste constituents apart from eggshell, which can be scaled-up for industrial application, optimizing processing factors to decrease void concentration, exploring machine learning methods to forecast long-term performance and performing comprehensive life-cycle evaluation to quantify environmental advantages.

4. Future scope

The future scope of this investigation could expand to analyzing additional biowaste components apart from eggshell such as calcium-

rich waste, food processing byproducts, or other agricultural residues in combination with the known inorganic particles. This extension would improve the sustainability profile while possibly identifying synergistic impacts between ceramic particles and waste materials. Also, scale-up investigation observing property consistency and manufacturability under industrial scales would be beneficial to bridge the laboratory to production gap, supplemented by long-term durability analysis under different environmental conditions. Further improvement of processing factors and manufacturing method could enhance filler distribution and decrease void concentration through alternative techniques like compression or resin transfer molding. Advanced machine learning methodology could be expanded to generate comprehensive predictive model which project composite characteristics with minimum experimental values, comprising the impacts of aging, cyclic loading and environmental exposure. This computational method integrated with comprehensive life cycle analysis would allow rapid optimization of formulation for particular industry requirement where measuring the environmental advantages of these waste-reinforced composites [181–183].

Authors' contributions

All authors equally contributed to the conception, experimental design, carrying out measurements, and manuscript composition.

Data and code availability

Not applicable.

Ethics approval

The authors hereby state that the present work is in compliance with the ethical standards.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors gratefully acknowledge the funding of the Deanship of Graduate Studies and Scientific Research, Jazan University, Saudi Arabia, through project number: (RG24-S0219).

References

- [1] Ponsoni LV, de Almeida MK, Madeira K, Militão GP, Zimmermann MVG. Statistical analysis of the substitution of inorganic fibers and fillers with vegetable fibers and fillers in polystyrene composites. *J Appl Polym Sci* 2024;141. <https://doi.org/10.1002/app.55497>.
- [2] Chandran AJ, Rangappa SM, Suyambulingam I, Siengchin S. Micro/nano fillers for value-added polymer composites: a comprehensive review. *J Vinyl Addit Technol* 2024;30:1083–123. <https://doi.org/10.1002/vnl.22106>.
- [3] Somasundaram R, Senthamaikannan P, Suyambulingam I, Kanthababu M, Madhu GM, Siengchin S. Greening the composite industry: evaluating Lantana camara Verbenaceae fiber as a promising substitute for lightweight polymer matrix composites. *Ind Crops Prod* 2024;220:119265. <https://doi.org/10.1016/j.indcrop.2024.119265>.
- [4] Mohammed L, Ansari MNM, Pua G, Jawaid M, Islam MS. A review on natural fiber reinforced polymer composite and its applications. *Int J Polym Sci* 2015; 2015:1–15. <https://doi.org/10.1155/2015/243947>.
- [5] Natural fibers and their composites. *Tribology of natural fiber polymer composites*. Elsevier; 2008. p. 1–58. <https://doi.org/10.1533/9781845695057.1>.
- [6] Sinha RK, Sridhar K, Purohit R, Malviya RK. Effect of nano SiO₂ on properties of natural fiber reinforced epoxy hybrid composite: a review. *Mater Today Proc* 2020;26:3183–6. <https://doi.org/10.1016/j.matpr.2020.02.657>.
- [7] Repon MdR, Islam T, Islam T, Alim MdA. Manufacture of polymer composites from plant fibers. *Plant biomass derived materials*. Wiley; 2024. p. 363–88. <https://doi.org/10.1002/9783527839032.ch14>.
- [8] Akter M, Uddin MdH, Anik HR. Plant fiber-reinforced polymer composites: a review on modification, fabrication, properties, and applications. *Polym Bull* 2024;81:1–85. <https://doi.org/10.1007/s00289-023-04733-5>.
- [9] Andoko A, Diharjo K, Soedarsono JW, Gapsari F, Rangappa SM, Siengchin S. Influence of various chemical treatments on mechanical, thermal and morphological characteristics of Cordia dichotoma fiber. *Case Stud Chem Environ Eng* 2024;10:100972. <https://doi.org/10.1016/j.csee.2024.100972>.
- [10] Diharjo K, Gapsari F, Andoko A, Septiari R, Rangappa SM, Siengchin S. Optimization of nano cellulose extraction from timoho fiber using response surface methodology (RSM). *Biomass Convers Biorefin* 2024;14:25557–67. <https://doi.org/10.1007/s13399-023-04551-9>.
- [11] Gapsari F, Putri TM, Rukmana W, Juliano H, Sulaiman AM, Dewi FGU, et al. Isolation and characterization of *Muntingia calabura* cellulose nanofibers. *J Nat Fibers* 2023;20. <https://doi.org/10.1080/15440478.2022.2156018>.
- [12] Andoko A, Gapsari F, Diharjo K, M RS, Siengchin S. Isolation of microcellulose from timoho fiber using the process of delignification and maceration: evaluation of physical, chemical, structural, and thermal properties. *Int J Biol Macromol* 2023;224:48–54. <https://doi.org/10.1016/j.ijbiomac.2022.10.225>.
- [13] Zmitrowicz A. Models of kinematics dependent anisotropic and heterogeneous friction. *Int J Solid Struct* 2006;43:4407–51. <https://doi.org/10.1016/j.ijsolstr.2005.07.001>.
- [14] Nipu SMA, Rahman MZ, Alam SS, Mechanical Dev B. Thermal and morphological characterization of graphene/Al₂O₃-reinforced epoxy hybrid nanocomposites. *Macromol Mater Eng* 2024. <https://doi.org/10.1002/mame.202400180>.
- [15] Thangarasu S, Shalu Palanisamy G, Sadhasivam S, Selvakumar K, Eswar Neerugatti KR, et al. Deciphering the role of 2D graphene oxide nanofillers in polymer membranes for vanadium redox flow batteries. *J Mater Chem A Mater* 2024;12:11176–234. <https://doi.org/10.1039/D4TA00001C>.
- [16] de Mendonça Neuba L, Felipe Pereira Junio R, Silva DS, Palmeira AA, Tavares S, Lazarus B, et al. Thermal, morphological, and structural characterization of Seven-Islands-Sedge (*Cyperus malaccensis*) fiber with graphene oxide as a coupling agent for future polymer composite reinforcement. *J Nat Fibers* 2024;21. <https://doi.org/10.1080/15440478.2024.2392284>.
- [17] Han X, Zhao Y, Sun J, Li Y, Zhang J, Hao Y. Effect of graphene oxide addition on the interlaminar shear property of carbon fiber-reinforced epoxy composites. *N Carbon Mater* 2017;32:48–55. [https://doi.org/10.1016/S1872-5805\(17\)60107-0](https://doi.org/10.1016/S1872-5805(17)60107-0).
- [18] Felix Sahayaraj A, M TS, Sasi Kumar M, Sathish S, Gokulkumar S, Jenish I, et al. Fire retardant potential of natural fiber reinforced polymer composites: a review. *Polymer-Plast Technol Mater* 2024;63:771–97. <https://doi.org/10.1080/25740881.2024.2303608>.
- [19] Thomas MG, Abraham E, Jyotishkumar P, Maria HJ, Pothen LA, Thomas S. Nanocelluloses from jute fibers and their nanocomposites with natural rubber: preparation and characterization. *Int J Biol Macromol* 2015;81:768–77. <https://doi.org/10.1016/j.ijbiomac.2015.08.053>.
- [20] Jordan J, Jacob KI, Tannenbaum R, Sharaf MA, Jasiuk I. Experimental trends in polymer nanocomposites—a review. *Mater Sci Eng, A* 2005;393:1–11. <https://doi.org/10.1016/j.msea.2004.09.044>.
- [21] Saba N, Tahir P, Jawaid M. A review on potentiality of nano filler/natural fiber filled polymer hybrid composites. *Polymers (Basel)* 2014;6:2247–73. <https://doi.org/10.3390/polym6082247>.
- [22] Ruiz-Pérez L, Royston GJ, Fairclough JPA, Ryan AJ. Toughening by nanostructure. *Polymer (Guildf)* 2008;49:4475–88. <https://doi.org/10.1016/j.polymer.2008.07.048>.
- [23] Liu Y, Yang J, Bai L, Qu D, Liu Z, Wang Z, et al. Mechanistic insights of nanofillers tuning polymer microstructures for enhanced separation performance of nanofiltration membrane. *Sep Purif Technol* 2024;341:126847. <https://doi.org/10.1016/j.seppur.2024.126847>.
- [24] Shahrajabian H, Vaezzadeh H. The nano-clay effect on the improvement of the thermal, flammability, and mechanical behavior of epoxy/glass fiber/ATH hybrid composites. *J Compos Mater* 2024. <https://doi.org/10.1177/00219983241270992>.
- [25] Mohsenzadeh R, Majidi H, Soltanzadeh M, Shelesh-Nezhad K. Wear and failure of polyoxymethylene/calcium carbonate nanocomposite gears. *Proc IME J J Eng Tribol* 2020;234:811–20. <https://doi.org/10.1177/1350650119867530>.
- [26] Feleke K, Ganesh T, Beri H, Murtaza A. Preparation and characterization of crystalline nanocellulose (CNC) from linseed straw fibers: as a potential alternative source of nanofillers in polymer composites. *J Eng Fiber Fabr* 2024;19. <https://doi.org/10.1177/15589250241265709>.
- [27] Singh K, Ohlan A, Saini P, Dhawan SK. Poly (3,4-ethylenedioxythiophene) γ -Fe₂O₃ polymer composite—super paramagnetic behavior and variable range hopping 1D conduction mechanism—synthesis and characterization. *Polym Adv Technol* 2008;19:229–36. <https://doi.org/10.1002/pat.1003>.
- [28] Chaudhary V, Ahmad F. A review on plant fiber reinforced thermoset polymers for structural and frictional composites. *Polym Test* 2020;91:106792. <https://doi.org/10.1016/j.polymertesting.2020.106792>.
- [29] Olson E, Li Y, Lin F-Y, Miller A, Liu F, Tsyrenova A, et al. Thin biobased transparent UV-blocking coating enabled by nanoparticle self-assembly. *ACS Appl Mater Interfaces* 2019;11:24552–9. <https://doi.org/10.1021/acsami.9b05383>.
- [30] Jin S-E, Jin H-E. Antimicrobial activity of zinc oxide nano/microparticles and their combinations against pathogenic microorganisms for biomedical applications: from physicochemical characteristics to pharmacological aspects. *Nanomaterials* 2021;11:263. <https://doi.org/10.3390/nano11020263>.
- [31] Barros AB de S, Farias R de F, Siqueira DD, Luna CBB, Araújo EM, Rabello MS, et al. The effect of ZnO on the failure of PET by environmental stress cracking. *Materials* 2020;13:2844. <https://doi.org/10.3390/ma13122844>.

- [32] Kiruthika AV. A review on physico-mechanical properties of bast fibre reinforced polymer composites. *J Build Eng* 2017;9:91–9. <https://doi.org/10.1016/j.jobe.2016.12.003>.
- [33] Kiruthika AV. A review of leaf fiber reinforced polymer composites. *J Eng Appl Sci* 2024;71:24. <https://doi.org/10.1186/s44147-024-00365-2>.
- [34] Pujar V, Devarajiah RM, Suresha B, Bharat V. A review on mechanical and wear properties of fiber-reinforced thermoset composites with ceramic and lubricating fillers. *Mater Today Proc* 2021;46:7701–10. <https://doi.org/10.1016/j.matpr.2021.02.214>.
- [35] Zulantoni Smaradhana DF. The manufacturing process and corrosion properties of magnesium matrix composites: a review. *Mech Explor Mater Innov* 2024;1: 7–19. <https://doi.org/10.21776/ub.memi.2024.001.01.2>.
- [36] Alemi-Ardakani M, Milani AS, Yannacopoulos S, Shokouhi G. On the effect of subjective, objective and combinative weighting in multiple criteria decision making: a case study on impact optimization of composites. *Expert Syst Appl* 2016;46:426–38. <https://doi.org/10.1016/j.eswa.2015.11.003>.
- [37] Singh T. A hybrid multiple-criteria decision-making approach for selecting optimal automotive brake friction composite. *Mater Des Process Commun* 2021; 3. <https://doi.org/10.1002/mdp2.266>.
- [38] Singh T. Utilization of cement bypass dust in the development of sustainable automotive brake friction composite materials. *Arab J Chem* 2021;14:103324. <https://doi.org/10.1016/j.arabjc.2021.103324>.
- [39] Chinchalikar S, Shinde S, Shaikh A, Gaikwad V, Ambhore NH. Multi-objective optimization of FDM using hybrid genetic algorithm-based multi-criteria decision-making (MCDM) techniques. *J Inst Eng: Ser D* 2024;105:49–63. <https://doi.org/10.1007/s40033-023-00459-w>.
- [40] Heddar MIE, Mehdi B, Matougui N, Tahan SA, Jahazi M. Adaptable multi-objective optimization framework: application to metal additive manufacturing. *Int J Adv Des Manuf Technol* 2024;132:1897–914. <https://doi.org/10.1007/s00170-024-13489-9>.
- [41] Oroujzadeh M, Nikouei MA, Mehdipour-Ataei S, Amiri M. Materials selection for choosing the best composite blend polymeric membrane for hydrogen/oxygen proton exchange membrane fuel cell. *J Power Sources* 2022;538:231566. <https://doi.org/10.1016/j.jpowsour.2022.231566>.
- [42] Singh T. Optimum design based on fabricated natural fiber reinforced automotive brake friction composites using hybrid CRITIC-MEW approach. *J Mater Res Technol* 2021;14:81–92. <https://doi.org/10.1016/j.jmrt.2021.06.051>.
- [43] Soni A, Chakraborty S, Kumar Das P, Kumar Saha A. Materials selection of reinforced sustainable composites by recycling waste plastics and agro-waste: an integrated multi-criteria decision making approach. *Constr Build Mater* 2022; 348:128608. <https://doi.org/10.1016/j.conbuildmat.2022.128608>.
- [44] Singh T, Pattnaik P, Kumar SR, Fekete G, Dogosy G, Lendvai L. Optimization on physicochemical and wear properties of wood waste filled poly(lactic acid) biocomposites using integrated entropy-simple additive weighting approach. *S Afr J Chem Eng* 2022;41:193–202. <https://doi.org/10.1016/j.sajce.2022.06.008>.
- [45] Maniya K, Bhatt MG. A selection of material using a novel type decision-making method: preference selection index method. *Mater Des* 2010;31:1785–9. <https://doi.org/10.1016/j.matdes.2009.11.020>.
- [46] Rampal Zafar S. Experimental analysis and multi-criteria optimization of process parameters in microwave drilling of kenaf-reinforced polypropylene composites. *J Eng Res* 2024. <https://doi.org/10.1016/j.jer.2024.06.001>.
- [47] Mahajan A, Singh I, Arora N. An integrated multi-criteria decision-making framework for the selection of sustainable biodegradable polymer for food packaging applications. *Environ Dev Sustain* 2023;26:8399–420. <https://doi.org/10.1007/s10668-023-03052-z>.
- [48] Rajan K. Materials informatics. *Mater Today* 2005;8:38–45. [https://doi.org/10.1016/S1369-7021\(05\)71123-8](https://doi.org/10.1016/S1369-7021(05)71123-8).
- [49] Ramprasad R, Batra R, Piliang G, Mannodi-Kanakkithodi A, Kim C. Machine learning in materials informatics: recent applications and prospects. *npj Comput Mater* 2017;3:54. <https://doi.org/10.1038/s41524-017-0056-5>.
- [50] Yin B, Huang J, Sun W. Peridynamics-fueled convolutional neural network for predicting mechanical constitutive behaviors of fiber reinforced composites. *Comput Methods Appl Mech Eng* 2024;431:117309. <https://doi.org/10.1016/j.cma.2024.117309>.
- [51] Yang C, Kim Y, Ryu S, Gu GX. Prediction of composite microstructure stress-strain curves using convolutional neural networks. *Mater Des* 2020;189:108509. <https://doi.org/10.1016/j.matdes.2020.108509>.
- [52] Kim D-W, Lim JH, Lee S. Prediction and validation of the transverse mechanical behavior of unidirectional composites considering interfacial debonding through convolutional neural networks. *Compos B Eng* 2021;225:109314. <https://doi.org/10.1016/j.compositesb.2021.109314>.
- [53] Gallagher B, Rever M, Loveland D, Mundhenk TN, Beauchamp B, Robertson E, et al. Predicting compressive strength of consolidated molecular solids using computer vision and deep learning. *Mater Des* 2020;190:108541. <https://doi.org/10.1016/j.matdes.2020.108541>.
- [54] Melro A, Camanho P, Pinho S. Generation of random distribution of fibres in long-fibre reinforced composites. *Compos Sci Technol* 2008;68:2092–102. <https://doi.org/10.1016/j.compscitech.2008.03.013>.
- [55] Swaminathan S, Ghosh S, Pagano NJ. Statistically equivalent representative volume elements for unidirectional composite microstructures: Part I - without damage. *J Compos Mater* 2006;40:583–604. <https://doi.org/10.1177/0021998305055273>.
- [56] Yabansu YC, Steinmetz P, Hötzer J, Kalidindi SR, Nestler B. Extraction of reduced-order process-structure linkages from phase-field simulations. *Acta Mater* 2017; 124:182–94. <https://doi.org/10.1016/j.actamat.2016.10.071>.
- [57] Paulson NH, Priddy MW, McDowell DL, Kalidindi SR. Data-driven reduced-order models for rank-ordering the high cycle fatigue performance of polycrystalline microstructures. *Mater Des* 2018;154:170–83. <https://doi.org/10.1016/j.matdes.2018.05.009>.
- [58] Ford E, Maneparambil K, Rajan S, Neithalath N. Machine learning-based accelerated property prediction of two-phase materials using microstructural descriptors and finite element analysis. *Comput Mater Sci* 2021;191:110328. <https://doi.org/10.1016/j.commatsci.2021.110328>.
- [59] Ammasai Sengodan G. Prediction of two-phase composite microstructure properties through deep learning of reduced dimensional structure-response data. *Compos B Eng* 2021;225:109282. <https://doi.org/10.1016/j.compositesb.2021.109282>.
- [60] Li A, Lyu Y, Yang B, Yang D. Novel repair of bolted composite joints using 3D printed continuous fibre patches with custom fibre paths. *Compos B Eng* 2025; 295:112212. <https://doi.org/10.1016/j.compositesb.2025.112212>.
- [61] Suhas K, Murthy BRN, Hiremath A. Fabrication and characterization approach to enhance the mechanical performance of zirconia-coated <sc>Multi-walled carbon nanotubes</sc> reinforced <sc>High density polyethylene</sc> composites. *Polym Compos* 2025. <https://doi.org/10.1002/pc.29574>.
- [62] Arriaga-Dávila J, Rosero-Arias C, Jonker D, Córdova-Castro M, Zscheile J, Kirchner R, et al. From single to multi-material 3D printing of glass-ceramics for micro-optics. *Small Methods* 2025. <https://doi.org/10.1002/smt.202401809>.
- [63] Yadav SJ, Thakor S, Shah DV, Venkatachalam S, Vaja CR. Synthesis and characterization of iron selenide-epoxy composites for enhanced optical, dielectric and EMI shielding applications. *J Polym Res* 2025;32:77. <https://doi.org/10.1007/s10965-025-04302-6>.
- [64] Panchal P, Shingala B, Thakor S, Jain P, Vaja CR, Joshi A, et al. XGBoost regression analysis of dielectric properties of epoxy resin with inorganic hybrid nanofillers. *J Macromol Sci, Part B* 2025;64:347–63. <https://doi.org/10.1080/0022348.2024.2347746>.
- [65] Zhang J, Zhou Z, Zhang C, Fu Z, Zhou S, Shao J, et al. Carbon nanotubes-intervened interface design of quartz fiber/polyurethane composite fibers towards improved mechanical properties. *Compos Sci Technol* 2025;262:111065. <https://doi.org/10.1016/j.compscitech.2025.111065>.
- [66] Arpitha GR, Mohit H, Madhu P, Verma A. Effect of sugarcane bagasse and alumina reinforcements on physical, mechanical, and thermal characteristics of epoxy composites using artificial neural networks and response surface methodology. *Biomass Convers Biorefin* 2024;14:12539–57. <https://doi.org/10.1007/s13399-023-03886-7>.
- [67] Pattanaik A, Mukharjee M, Mishra SC. Effect of environmental aging conditions on the properties of fly ash filled epoxy composites. *Adv Compos Mater* 2020;29: 1–30. <https://doi.org/10.1080/09243046.2019.1610930>.
- [68] Yang L, Liu X, Wu Z, Wang R. Effects of triangle-shape fiber on the transverse mechanical properties of unidirectional carbon fiber reinforced plastics. *Compos Struct* 2016;152:617–25. <https://doi.org/10.1016/j.compstruct.2016.05.065>.
- [69] Salvatori D, Caglar B, Teixidó H, Michaud V. Permeability and capillary effects in a channel-wise non-crimp fabric. *Compos Part A Appl Sci Manuf* 2018;108:41–52. <https://doi.org/10.1016/j.compositesa.2018.02.015>.
- [70] Han K. Measurements of the permeability of fiber preforms and applications. *Compos Sci Technol* 2000;60:2435–41. [https://doi.org/10.1016/S0266-3538\(00\)00037-3](https://doi.org/10.1016/S0266-3538(00)00037-3).
- [71] Rao HJ, Singh S, Singh N, Ramulu PJ, Santos TF, Santos CM, et al. Enhancing mechanical performance and water resistance of Careya-Banana fiber epoxy hybrid composites through PLA coating and alkali treatment. *J Mater Res Technol* 2024;32:4304–15. <https://doi.org/10.1016/j.jmrt.2024.08.190>.
- [72] Balaji Ayyanar C, Kumar R, Helaili S, B G, Rinasuba V, Nalini HE, et al. Experimental and numerical analysis of natural fillers loaded and E-glass reinforced epoxy sandwich composites. *J Mater Res Technol* 2024;32:1235–44. <https://doi.org/10.1016/j.jmrt.2024.07.142>.
- [73] Paruthi S, Khan AH, Kumar A, Kumar F, Hasan MA, Magbool HM, et al. Sustainable cement replacement using waste eggshells: a review on mechanical properties of eggshell concrete and strength prediction using artificial neural network. *Case Stud Constr Mater* 2023;18. <https://doi.org/10.1016/j.cscm.2023.e02160>.
- [74] Bhagwan D, Agarwal LJBKC. Analysis and performance of fiber composites. 4th, vol. 1. Wiley; 2017.
- [75] Laachachi A, Ferriol M, Cochez M, Lopez Cuesta J-M, Ruch D. A comparison of the role of boehmite (AlOOH) and alumina (Al₂O₃) in the thermal stability and flammability of poly(methyl methacrylate). *Polym Degrad Stabil* 2009;94: 1373–8. <https://doi.org/10.1016/j.polymdegradstab.2009.05.014>.
- [76] Cheng L, Huang C, Yang Y, Li Y, Meng Y, Li Y, et al. Preparation and combustion performance of B/PVDF/Al composite microspheres. *Propellants, Explos Pyrotech* 2020;45:657–64. <https://doi.org/10.1002/prep.201900232>.
- [77] Li X, Yang H, Li Y. Characterization of thermal reaction of aluminum/copper (II) oxide/poly(tetrafluoroethylene) nanocomposite by thermogravimetric analysis, differential scanning calorimetry, mass spectrometry and X-ray diffraction. *Thermochim Acta* 2015;621:68–73. <https://doi.org/10.1016/j.tca.2015.10.012>.
- [78] Gu A, Liang G. Thermal degradation behaviour and kinetic analysis of epoxy/montmorillonite nanocomposites. *Polym Degrad Stabil* 2003;80:383–91. [https://doi.org/10.1016/S0141-3910\(03\)00026-0](https://doi.org/10.1016/S0141-3910(03)00026-0).
- [79] Yousef S, Kiminaité I, Eimontas J, Striugas N, Abdelnaby MA. Catalytic pyrolysis kinetic behaviour of glass fibre-reinforced epoxy resin composites over ZSM-5 zeolite catalyst. *Fuel* 2022;315:123235. <https://doi.org/10.1016/j.fuel.2022.123235>.
- [80] Yousef S, Eimontas J, Striugas N, Praspaliauskas M, Abdelnaby MA. Pyrolysis kinetic behaviour of glass fibre-reinforced epoxy resin composites using linear

- and nonlinear isoconversional methods. *Polymers* (Basel) 2021;13:1543. <https://doi.org/10.3390/polym13101543>.
- [81] Bignotti F, Pandini S, Baldi F, De Santis R. Effect of the resin/hardener ratio on curing, structure and glass transition temperature of nanofilled epoxies. *Polym Compos* 2011;32:1034–48. <https://doi.org/10.1002/pc.21120>.
- [82] Zaharescu T, Bumbac M, Nicoliescu CM. Stability effects of added biomass on microalgae styrene-butadiene-styrene composites. *J Therm Anal Calorim* 2024; 149:10793–805. <https://doi.org/10.1007/s10973-024-13419-7>.
- [83] Zhu R, Liu H, Zhang J. Compatibilizing effects of maleated poly(lactic acid) (PLA) on properties of PLA/soy protein composites. *Ind Eng Chem Res* 2012;51: 7786–92. <https://doi.org/10.1021/ie300118x>.
- [84] Gapsari F, Utaminigum F, Lai CW, Anam K, Sulaiman AM, Haidar MF, et al. A convolutional neural network-VGG16 method for corrosion inhibition of 304SS in sulfuric acid solution by timoho leaf extract. *J Mater Res Technol* 2024;30: 1116–27. <https://doi.org/10.1016/j.jmrt.2024.03.156>.
- [85] Andoko A, Gapsari F, Harmayanti A, Sulaiman AM, Prasetya R, Putri N, et al. Enhanced performance of epoxy composites: a study on Walikukun-glass fiber hybrid reinforcement for automotive applications. *Int J Lightweight Mater Manuf* 2025. <https://doi.org/10.1016/j.ijlmm.2025.03.003>.
- [86] Ornaghi HL, Monticeli FM, Neves RM, Zattera AJ, Cioffi MOH, Voorwald HJC. Effect of stacking sequence and porosity on creep behavior of glass/epoxy and carbon/epoxy hybrid laminate composites. *Compos Commun* 2020;19:210–9. <https://doi.org/10.1016/j.coco.2020.04.006>.
- [87] Alamri H, Low IM. Mechanical properties and water absorption behaviour of recycled cellulose fibre reinforced epoxy composites. *Polym Test* 2012;31:620–8. <https://doi.org/10.1016/j.polymertesting.2012.04.002>.
- [88] Muñoz E, García-Manrique JA. Water absorption behaviour and its effect on the mechanical properties of flax fibre reinforced bioepoxy composites. *Int J Polym Sci* 2015;2015:1–10. <https://doi.org/10.1155/2015/390275>.
- [89] Yang J, Huang D, Feng Z, He H, Jin M. Effect of aerogel incorporation on water content distribution and capillary water absorption in pore size intervals of cementitious composites. *J Mater Sci* 2024;59:13756–71. <https://doi.org/10.1007/s10853-024-09879-9>.
- [90] Ng LF, Yahya MY, Muthukumar C, Parameswaranpillai J, Ma Q, Muhammad Asyraf MR, et al. Mechanical characterization, water absorption, and thickness swelling of lightweight pineapple leaf/ramie fabric-reinforced polypropylene hybrid composites. *Polymers* (Basel) 2024;16:1847. <https://doi.org/10.3390/polym16131847>.
- [91] Belaadi A, Lekrine A, Boumaaza M, Alshahrani H, Bourchak M, A Juhany K, et al. Water uptake of HDPE reinforced with *Washingtonia* fibre biocomposites: mathematical modelling using artificial neural network, response surface methodology and genetic algorithm. *Adv Mater Process Technol* 2023;1–24. <https://doi.org/10.1080/2374068X.2023.2198828>.
- [92] Yuan Q, Xie Z, Tian Y, Sun LM, Yang C, Chi H, et al. The driving force of water absorption in cementitious materials: an analysis of surface-free energies and pore structure. *J Mater Res Technol* 2024;30:4714–24. <https://doi.org/10.1016/j.jmrt.2024.04.167>.
- [93] Kou Y, Yi M, Li X, She Y, Li Q, Shao L, et al. Preparation of lightweight porous slag-based geopolymer highly hydrophobic materials for efficient oil-water separation. *Constr Build Mater* 2024;439:137296. <https://doi.org/10.1016/j.conbuildmat.2024.137296>.
- [94] Rathod VT, Kumar JS, Jain A. Polymer and ceramic nanocomposites for aerospace applications. *Appl Nanosci* 2017;7:519–48. <https://doi.org/10.1007/s13204-017-0592-9>.
- [95] Puliyalil H, Filipić G, Cvelbar U. Selective plasma etching of polymers and polymer matrix composites. Non-thermal plasma technology for polymeric materials. Elsevier; 2019. p. 241–59. <https://doi.org/10.1016/B978-0-12-813152-7.00009-3>.
- [96] Rafi EA, Aziz M Bin, Khan MdTR, Haque MR, Hasan M, Gafur MA, et al. Effect of Sesbania grandiflora stem fiber reinforcement on mechanical, chemical, thermal, and physical properties of vinyl ester material. *Biomass Convers Biorefin* 2024. <https://doi.org/10.1007/s13399-024-06105-z>.
- [97] Sabu M, Ruban YJV, Ginil MS. Effect of organoclay on contact angle, thermal and mechanical properties of filled epoxy composites. *Mater Today Proc* 2020;41: 549–56. <https://doi.org/10.1016/j.matpr.2020.05.248>. Elsevier Ltd.
- [98] Borrás A, López C, Rico V, Gracia F, González-Elipe AR, Richter E, et al. Effect of visible and UV illumination on the water contact angle of TiO₂ thin films with incorporated nitrogen. *J Phys Chem C* 2007;111:1801–8. <https://doi.org/10.1021/jp065392w>.
- [99] Yi F, Guo ZX, Zhang LX, Yu J, Li Q. Soluble eggshell membrane protein: preparation, characterization and biocompatibility. *Biomaterials* 2004;25: 4591–9. <https://doi.org/10.1016/j.biomaterials.2003.11.052>.
- [100] Megias-Alguacil D, Tervoort E, Cattin C, Gauckler LJ. Contact angle and adsorption behavior of carboxylic acids on α -Al₂O₃ surfaces. *J Colloid Interface Sci* 2011;353:512–8. <https://doi.org/10.1016/j.jcis.2010.09.087>.
- [101] Sever K, Seki Y, Tavman IH, Erkan G, Cecen V. The structure of γ -glycidioxypropyltrimethoxysilane on glass fiber surfaces: characterization by FTIR, SEM, and contact angle measurements. *Polym Compos* 2009;30:550–8. <https://doi.org/10.1002/pc.20585>.
- [102] Vovk M, Planinšek O, German Ilić I, Šernek M. Characterization of industrial aluminum trihydrate-filled poly(methyl methacrylate) composite powder. *J Adhes Sci Technol* 2019;33:2517–34. <https://doi.org/10.1080/01694243.2019.1654209>.
- [103] Bismarck A, Boccaccini AR, Egia-Ajuriagoaskoa E, Ulsenberg', Leutbecher T. Surface characterization of glass fibers made from silicate waste: zeta-potential and contact angle measurements. n.d.
- [104] Goss BGS, Birtwhistle' D, George GA. A comparison between the chemical effects of degradation and the physical and electrical performance of EPDM rubber. n.d.
- [105] Redón R, Vázquez-Olmos A, Mata-Zamora ME, Ordóñez-Medrano A, Rivera-Torres F, Saniger JM. Contact angle studies on anodic porous alumina. *J Colloid Interface Sci* 2005;287:664–70. <https://doi.org/10.1016/j.jcis.2005.02.036>.
- [106] Megias-Alguacil D, Tervoort E, Cattin C, Gauckler LJ. Contact angle and adsorption behavior of carboxylic acids on α -Al₂O₃ surfaces. *J Colloid Interface Sci* 2011;353:512–8. <https://doi.org/10.1016/j.jcis.2010.09.087>.
- [107] Rupp F, Axmann D, Ziegler C, Geis-Gerstorf J. Adsorption/desorption phenomena on pure and Teflon AF-coated titania surfaces studied by dynamic contact angle analysis. 2002.
- [108] Kumar R, Kumar K, Bhowmik S. Mechanical characterization and quantification of tensile, fracture and viscoelastic characteristics of wood filler reinforced epoxy composite. *Wood Sci Technol* 2018;52:677–99. <https://doi.org/10.1007/s00226-018-0995-0>.
- [109] Zhang H. Effect of a novel coupling agent, alkyl ketene dimer, on the mechanical properties of wood-plastic composites. *Mater Des* 2014;59:130–4. <https://doi.org/10.1016/j.matdes.2014.02.048>.
- [110] Donmez Cavdar A, Mengelöglü F, Karakus K. Effect of boric acid and borax on mechanical, fire and thermal properties of wood flour filled high density polyethylene composites. *Measurement* 2015;60:6–12. <https://doi.org/10.1016/j.measurement.2014.09.078>.
- [111] Anju R, Ramesan MT. Enhanced mechanical, electrical, thermal, and optical properties of poly(methyl methacrylate)/copper oxide nanocomposites for flexible optoelectronic devices via in-situ polymerization technique. *Polym Compos* 2024;45:5360–71. <https://doi.org/10.1002/pc.28132>.
- [112] Sliwa F, El Bounia N, Charrier F, Marin G, Malet F. Mechanical and interfacial properties of wood and bio-based thermoplastic composite. *Compos Sci Technol* 2012;72:1733–40. <https://doi.org/10.1016/j.compscitech.2012.07.002>.
- [113] Ndiaye D, Gueye M, Diop B. Characterization, physical and mechanical properties of polypropylene/wood-flour composites. *Arabian J Sci Eng* 2013;38:59–68. <https://doi.org/10.1007/s13369-012-0407-y>.
- [114] Sepet H, Tarakcioglu N, Misra R. Investigation of mechanical, thermal and surface properties of nanoclay/HDPE nanocomposites produced industrially by melt mixing approach. *J Compos Mater* 2016;50:3105–16. <https://doi.org/10.1177/0021998315615653>.
- [115] Gapsari F, Wijatmiko I, Andoko A, Diharjo K, Mavinkere Rangappa S, Siengchin S. Modification on fiber from alkali treatment and AESO coating to enhance UV-light and water absorption resistance in Kapok fiber reinforced polyester composites. *J Nat Fibers* 2024;21. <https://doi.org/10.1080/15440478.2024.2383970>.
- [116] Mohan TP, Kanny K. Mechanical and thermal properties of nanoclay-treated banana fibers. *J Nat Fibers* 2017;14:718–26. <https://doi.org/10.1080/15440478.2016.1277819>.
- [117] Floros M, Hojabri L, Abraham E, Jose J, Thomas S, Pothan L, et al. Enhancement of thermal stability, strength and extensibility of lipid-based polyurethanes with cellulose-based nanofibers. *Polym Degrad Stab* 2012;97:1970–8. <https://doi.org/10.1016/j.polymdegradstab.2012.02.016>.
- [118] Deepa B, Abraham E, Cherian BM, Bismarck A, Blaker JJ, Pothan LA, et al. Structure, morphology and thermal characteristics of banana nano fibers obtained by steam explosion. *Bioresour Technol* 2011;102:1988–97. <https://doi.org/10.1016/j.biortech.2010.09.030>.
- [119] Cordeiro N, Faria M, Abraham E, Pothan LA. Assessment of the changes in the cellulosic surface of micro and nano banana fibres due to saponin treatment. *Carbohydr Polym* 2013;98:1065–71. <https://doi.org/10.1016/j.carbpol.2013.07.002>.
- [120] Mohan TP, Kanny K. Compressive characteristics of unmodified and nanoclay treated banana fiber reinforced epoxy composite cylinders. *Compos B Eng* 2019; 169:118–25. <https://doi.org/10.1016/j.compositesb.2019.03.071>.
- [121] Yusuf IT, Jimoh YA, Salami WA. An appropriate relationship between flexural strength and compressive strength of palm kernel shell concrete. *Alex Eng J* 2016; 55:1553–62. <https://doi.org/10.1016/j.aej.2016.04.008>.
- [122] Üstün T, Saraloğlu Güler E. Enhancing carbon fiber reinforced aluminum laminates with cellulose paper interlayers: experimental characterization of tensile, flexural, and interlaminar fracture toughness. *Compos Interfaces* 2024; 1–21. <https://doi.org/10.1080/09276440.2024.2331332>.
- [123] Mirmehdi SM, Zeinaly F, Dabbagh F. Date palm wood flour as filler of linear low-density polyethylene. *Compos B Eng* 2014;56:137–41. <https://doi.org/10.1016/j.compositesb.2013.08.008>.
- [124] Wang G, Yu M, Wang M, Liu X, Zhang H, He L. Study of interlaminar fracture toughness and flexural properties of carbon fibre composites reinforced with polyethersulfone/graphene oxide films. *Compos Commun* 2024;51:102051. <https://doi.org/10.1016/j.coco.2024.102051>.
- [125] Khan SU, Kim J-K. Improved interlaminar shear properties of multiscale carbon fiber composites with bucky paper interleaves made from carbon nanofibers. *Carbon N Y* 2012;50:5265–77. <https://doi.org/10.1016/j.carbon.2012.07.011>.
- [126] Zheng C, Tang Z, Li Y, Zhang S, Chen C, Cheng L. Compression and interlaminar shear behavior of steered laminates manufactured by automated fiber placement. *Polym Compos* 2024. <https://doi.org/10.1002/pc.28843>.
- [127] Hussain M, Nakahira A, Niihara K. Mechanical property improvement of carbon fiber reinforced epoxy composites by Al₂O₃ filler dispersion. *Mater Lett* 1996;26: 185–91. [https://doi.org/10.1016/0167-577X\(95\)00224-3](https://doi.org/10.1016/0167-577X(95)00224-3).
- [128] Hussain A, Podgursky V, Goljandin D, Antonov M. Industrial approach to circularity of polymer composites: processing, characterization, mechanical testing, and wear regression. *J Reinforc Plast Compos* 2024;43:456–72. <https://doi.org/10.1177/07316844231164563>.

- [129] Bazrgari D, Moztaazadeh F, Sabbagh-Alvani AA, Rasoulianboroujeni M, Tahriri M, Tayebi L. Mechanical properties and tribological performance of epoxy/Al₂O₃ nanocomposite. *Ceram Int* 2018;44:1220–4. <https://doi.org/10.1016/j.ceramint.2017.10.068>.
- [130] Jubier NJ, Al-Jorani KR, Ali AA, Al-Bayatay SA, Al-Uqaily RAH. Thermal degradation assessment, impact strength, and hardness of combination epoxy and polystyrene powder composite. *Kuwait J Sci* 2024;51:100271. <https://doi.org/10.1016/j.kjs.2024.100271>.
- [131] Raj T, Jain A, Raj A, Tyagi B, Sahai A, Swarup Sharma R. Infill strategies for improving the impact behavior of polymer composites utilizing statistical and thermal analysis. *Int J Polym Anal Char* 2024;29:109–26. <https://doi.org/10.1080/1023666X.2024.2319404>.
- [132] Younsri OM, Abdellatif MH, Bassioni G. Effect of Al₂O₃ nanoparticles on the mechanical and physical properties of epoxy composite. *Arabian J Sci Eng* 2018;43:1511–7. <https://doi.org/10.1007/s13369-017-2955-7>.
- [133] Koay SC, Husseinayah S, Osman H. Modified cocoa pod husk-filled polypropylene composites by using methacrylic acid. *Bioresources* 2013;8. <https://doi.org/10.15376/biores.8.3.3260-3275>.
- [134] Yağcı Ö, Eker Gümtüş B, Taşdemir M. Thermal, structural and dynamical mechanical properties of hollow glass sphere-reinforced polypropylene composites. *Polym Bull* 2021;78:3089–101. <https://doi.org/10.1007/s00289-020-03257-6>.
- [135] Król-Morkisz K, Pielichowska K. Thermal decomposition of polymer nanocomposites with functionalized nanoparticles. *Polymer composites with functionalized nanoparticles: synthesis, properties, and applications*. Elsevier; 2018. p. 405–35. <https://doi.org/10.1016/B978-0-12-814064-2.00013-5>.
- [136] Ray S, Cooney RP. Thermal degradation of polymer and polymer composites. *Handbook of environmental degradation of materials*. third ed. Elsevier Inc.; 2018. p. 185–206. <https://doi.org/10.1016/B978-0-323-52472-8.00009-5>.
- [137] Kumar S, Prasad L, Bijlwan PP, Yadav A. Thermogravimetric analysis of lignocellulosic leaf-based fiber-reinforced thermosets polymer composites: an overview. *Biomass Convers Biorefin* 2024;14:12673–98. <https://doi.org/10.1007/s13399-022-03332-0>.
- [138] Singh M, Ranolita T, Durga G, Kalra P. Silicon bearing organic–inorganic hybrid epoxy resin for the development of novel fiber reinforced polymer composites: a thermomechanical evaluation. *Silicon* 2024;16:5063–72. <https://doi.org/10.1007/s12633-024-03069-7>.
- [139] Kirubakaran R, Salunke DR, Pitchumani SV, Gopalan V, Sampath A. Development of hBN/natural fibres reinforced polymer composites using grey relation grade analysis for thermal and electrical applications. *Funct Compos Struct* 2024;6: 025002. <https://doi.org/10.1088/2631-6331/ad3edd>.
- [140] Ren PG, Si XH, Sun ZF, Ren F, Pei L, Hou SY. Synergistic effect of BN and MWCNT hybrid fillers on thermal conductivity and thermal stability of ultra-high-molecular-weight polyethylene composites with a segregated structure. *J Polym Res* 2016;23:1–11. <https://doi.org/10.1007/s10965-015-0908-y>.
- [141] Yang CR, Chen C Da, Cheng C, Shi WH, Chen PH, Teng TP. Synergistic effect of hybrid fillers/polydimethylsiloxane composites on enhancing thermal conductivity. *Case Stud Therm Eng* 2021;28. <https://doi.org/10.1016/j.csite.2021.101591>.
- [142] Wang X, Wang Z, Wang L, Hong K, Wei L, Wang H, et al. Mechanical and thermal properties of epoxy nanocomposites interfacial modified with styrene–butadiene–styrene grafted multi-walled carbon nanotubes. *Polym Compos* 2018;39. <https://doi.org/10.1002/pc.24388>.
- [143] Zhou TY, Tsui GCP, Liang JZ, Zou SY, Tang CY, Mišković-Stanković V. Thermal properties and thermal stability of PP/MWCNT composites. *Compos B Eng* 2016; 90:107–14. <https://doi.org/10.1016/j.compositesb.2015.12.013>.
- [144] Karvanis K, Rusnáková S, Krejčí O, Žaludek M. Preparation, thermal analysis, and mechanical properties of basalt fiber/epoxy composites. *Polymers (Basel)* 2020; 12:1785. <https://doi.org/10.3390/polym12081785>.
- [145] Lee S-E, Cho S, Lee Y-S. Mechanical and thermal properties of MWCNT-reinforced epoxy nanocomposites by vacuum assisted resin transfer molding. *Carbon Lett* 2014;15:32–7. <https://doi.org/10.5714/CL.2014.15.1.032>.
- [146] Cai W, Li Z, Mu X, He L, Zhou X, Guo W, et al. Barrier function of graphene for suppressing the smoke toxicity of polymer/black phosphorous nanocomposites with mechanism change. *J Hazard Mater* 2021;404. <https://doi.org/10.1016/j.jhazmat.2020.124106>.
- [147] Ren X, Zong R, Hu Y, Lo S, Stec AA, Hull TR. Investigation of thermal decomposition of polymer nanocomposites with different char residues. *Polym Adv Technol* 2015;26:1027–33. <https://doi.org/10.1002/pat.3527>.
- [148] Zeng S, Shen M, Yang L, Xue Y, Lu F, Chen S. Self-assembled montmorillonite–carbon nanotube for epoxy composites with superior mechanical and thermal properties. *Compos Sci Technol* 2018;162:131–9. <https://doi.org/10.1016/j.compscitech.2018.04.035>.
- [149] Wang J, Li X, Wu T, Chen Z, Zhao R, Wang Z, et al. Effects of crystallization and thermal degradation behavior on mechanical properties of fiber-reinforced composites during post-annealing. *Polym Compos* 2024. <https://doi.org/10.1002/pc.29397>.
- [150] Yao W, Pan S, Qiu Z. Crystallization behavior and mechanical property of biodegradable poly(butylene succinate-co-2-methyl succinate)/cellulose nanocrystals composites. *Polymers (Basel)* 2024;16:1735. <https://doi.org/10.3390/polym16121735>.
- [151] Guo X, Lian H, Qin J, Guo J, Chen X, Chu L. Effect of annealing treatment on impact toughness and crystallization behaviors of polypropylene/ethylene vinyl acetate copolymer/layered double hydroxide composites. *Polym Compos* 2024; 45:16193–204. <https://doi.org/10.1002/pc.28899>.
- [152] Yas MH, Mohammadi S, Astinchap B, Heshmati M. A comprehensive study on the thermo-mechanical properties of multi-walled carbon nanotube/epoxy nanocomposites. *J Compos Mater* 2016;50:2025–34. <https://doi.org/10.1177/0021998315598853>.
- [153] Semaan P, Zahran A, Schlapp-Hackl I, Tehrani-Bagha AR, Mustapha S. Enhanced mechanical properties of epoxy composites using cellulose micro- and nanocrystals. *J Appl Polym Sci* 2024;141. <https://doi.org/10.1002/app.55180>.
- [154] Srinivasa Perumal KP, Selvarajan L, Sekar SM, Natarajan E. Examining thermogravimetric response and morphological alterations in epoxy composites with hybrid ceramic fillers integration. *Mater Chem Phys* 2024;325:129755. <https://doi.org/10.1016/j.matchemphys.2024.129755>.
- [155] Putra FR, Pramana AS, Pamungkas BS, Mauliyansyah R, Suprpto W, Al-Hachimi AAJ. Effect of stirring time in induction furnace on density, porosity, efficiency, fluidity, and geometry of Al–Mg–Si castings. *Mech Explor Mater Innov* 2024;1:88–93. <https://doi.org/10.21776/ub.memi.2024.001.03.3>.
- [156] Anam K, Purnowidodo A, Ananta B, Kumar PKAV. The effect of load steps and initial crack length on stress distribution of fiber metal laminates. *Mech Explor Mater Innov* 2024;1:54–60. <https://doi.org/10.21776/ub.memi.2024.001.02.3>.
- [157] Siddiqui VU, Sapuan SM, Hassan MR. Innovative dispersion techniques of graphene nanoplatelets (GNPs) through mechanical stirring and ultrasonication: impact on morphological, mechanical, and thermal properties of epoxy nanocomposites. *Defence Technol* 2025;43:13–25. <https://doi.org/10.1016/j.dt.2024.04.018>.
- [158] Karagöz İ, Mutlu D, Çavuşoğlu A, Çelebi M, Ceylan Ö. A comprehensive study on the effect of small rates of walnut shell and talc fillers on the thermal, mechanical, and morphological properties of epoxy hybrid composites. *Biomass Convers Biorefin* 2024. <https://doi.org/10.1007/s13399-024-05660-9>.
- [159] S. C. O, D. N, Z. O, T. H. M, P. S. M, A. K. M. Enhancing mechanical and thermal properties of epoxy composites with fish scale-derived collagen reinforcement. *Adv Polym Technol* 2024;2024. <https://doi.org/10.1155/2024/8890654>.
- [160] Gojny FH, Schulte K. Functionalisation effect on the thermo-mechanical behaviour of multi-wall carbon nanotube/epoxy-composites. *Compos Sci Technol* 2004;64:2303–8. <https://doi.org/10.1016/j.compscitech.2004.01.024>.
- [161] Liu W, Hoa SV, Pugh M. Organoclay-modified high performance epoxy nanocomposites. *Compos Sci Technol* 2005;65:307–16. <https://doi.org/10.1016/j.compscitech.2004.07.012>.
- [162] Cheon J-W, Moon J-H, Lee O, Kim D-S, Lee DJ, Lee JH, et al. PES-grafted graphene core-shell filler for epoxy composites: significant improvements of thermal and mechanical properties at ultra-low loadings. *React Funct Polym* 2025;212:106225. <https://doi.org/10.1016/j.reactfunctpolym.2025.106225>.
- [163] Shi S, Zheng S, Zeng K, Yue L, Zhang X, Rao W, et al. In-situ functional modification of the inner and outer surfaces of mesoporous silica SBA-15 towards improving fire safety and thermal stability of epoxy resins. *Constr Build Mater* 2025;471:140672. <https://doi.org/10.1016/j.conbuildmat.2025.140672>.
- [164] Lu S, Jayaraman A. Machine learning for analyses and automation of structural characterization of polymer materials. *Prog Polym Sci* 2024;153:101828. <https://doi.org/10.1016/j.progpolymsci.2024.101828>.
- [165] Ördek B, Borgianni Y, Coatanea E. Machine learning-supported manufacturing: a review and directions for future research. *Prod Manuf Res* 2024;12. <https://doi.org/10.1080/21693277.2024.2326526>.
- [166] Rajendra Kumar P, Madhav Rao B, Maddireddy CKR, Thakor S, Vaja CR, Prakash K, et al. Dielectric characterization and machine learning-based predictions in polymer composites with mixed nanoparticles. *J Macromol Sci, Part B* 2024;1–15. <https://doi.org/10.1080/00222348.2024.2372522>.
- [167] Mahapatra SK, Satapathy A. Development of machine learning models for the prediction of erosion wear of hybrid composites. *Polym Compos* 2024;45: 7950–66. <https://doi.org/10.1002/pc.28315>.
- [168] Kiran MD, B RLY, Babbar A, Kumar R, H SSC, Shetty RP, et al. Tribological properties of CNT-filled epoxy-carbon fabric composites: optimization and modelling by machine learning. *J Mater Res Technol* 2024;28:2582–601. <https://doi.org/10.1016/j.jmrt.2023.12.175>.
- [169] Machello C, Aghabalaee Baghaei K, Bazli M, Hadigheh A, Rajabipour A, Arashpour M, et al. Tree-based machine learning approach to modelling tensile strength retention of Fibre Reinforced Polymer composites exposed to elevated temperatures. *Compos B Eng* 2024;270:111132. <https://doi.org/10.1016/j.compositesb.2023.111132>.
- [170] Wu Q, Liu Y, Dai Y, Guo H, Wang Y, Zhuang W. High-fidelity prediction of forming quality for self-piercing riveted joints in aluminum alloy based on machine learning. *Mater Today Commun* 2024;41:110319. <https://doi.org/10.1016/j.mtcomm.2024.110319>.
- [171] Kumar L, Goyal A, Garg S, Phanden RK. Predictive modeling of drilling machine performance for jute fiber-reinforced polymer composites using GA, TLBO, and GRA-based RSM approaches. *Int J Interact Des Manuf* 2024. <https://doi.org/10.1007/s12008-024-01892-1>.
- [172] Prakoso MW, Baroroh DN, Setswalo K. Water absorption rate of Kenaf fiber (KF)/hydroxiteapatite (HA) in simulated sea water. *Mech Explor Mater Innov* 2024;1: 35–41. <https://doi.org/10.21776/ub.memi.2024.001.01.5>.
- [173] Osa-uwagboe N, Udu AG, Ghalati MK, Silberschmidt VV, Aremu A, Dong H, et al. A machine learning-enabled prediction of damage properties for fiber-reinforced polymer composites under out-of-plane loading. *Eng Struct* 2024;308:117970. <https://doi.org/10.1016/j.engstruct.2024.117970>.
- [174] Rabby MM, Das PP, Rahman M, Vadlamudi V, Raihan R. Fast and accurate prediction of cure quality and mechanical performance in fiber-reinforced polymer composite using dielectric variables and machine learning. *Polym Compos* 2024;45:1810–25. <https://doi.org/10.1002/pc.27891>.

- [175] Pankaj Kant S, Jawalkar CS, Khatkar SK, Singh M, Jindal MK. Experimental investigation on mechanical performance and drilling behavior of hybrid polymer composites through statistical and machine learning approach. *Proc IME E J Process Mech Eng* 2024. <https://doi.org/10.1177/09544089231223022>.
- [176] Mohit H, Sanjay MR, Siengchin S, Kanaan B, Ali V, Alarifi IM, et al. Machine learning-based prediction of mechanical and thermal properties of nickel/cobalt/ferrous and dried leaves fiber-reinforced polymer hybrid composites. *Polym Compos* 2024;45:489–506. <https://doi.org/10.1002/pc.27793>.
- [177] Song L, Yin A, Chen S, Long Z. Adaptive residual convolutional neural network for compressive strength prediction of energetic materials using SEM images. *Phys Scr* 2024;99:056011. <https://doi.org/10.1088/1402-4896/ad3e37>.
- [178] Monson PM de C, Conceição Junior P de O, Lofrano Dotto FR, Aguiar PR de, Rodrigues AR, David GA. Structural damage classification in composite materials using the Wigner-Ville distribution and convolutional neural networks. *Mater Lett* 2024;369:136734. <https://doi.org/10.1016/j.matlet.2024.136734>.
- [179] Ding Z, Attar HR, Wang H, Liu H, Li N. Integrating convolutional neural network and constitutive model for rapid prediction of stress-strain curves in fibre reinforced polymers: a generalisable approach. *Mater Des* 2024;241:112849. <https://doi.org/10.1016/j.matdes.2024.112849>.
- [180] Demircan G, Kisa M, Ozen M, Aktas B. Surface-modified alumina nanoparticles-filled aramid fiber-reinforced epoxy nanocomposites: preparation and mechanical properties. *Iran Polym J* 2020;29:253–64. <https://doi.org/10.1007/s13726-020-00790-z>.
- [181] Naveen MR, Kamaraj L, Ponnarengan H. Sustainable optimization of compression molding parameters for high-performance titanium-based fiber metal laminates with graphene-enhanced <scp>K</scp> evlar/jute fibers. *Polym Compos* 2025. <https://doi.org/10.1002/pc.29582>.
- [182] Thanikodi S, Giri J, Saravanan R, Mohammad F. Eco-friendly alternatives to printed circuit boards: developing natural fiber polymer composites with optimized flammability and moisture resistance. *J Mater Sci Mater Electron* 2025; 36:368. <https://doi.org/10.1007/s10854-025-14441-7>.
- [183] Skosana SJ, Khoathane C, Malwela T. Driving towards sustainability: a review of natural fiber reinforced polymer composites for eco-friendly automotive light-weighting. *J Thermoplast Compos Mater* 2025;38:754–80. <https://doi.org/10.1177/08927057241254324>.
- [184] Maes S, Badi N, Winne JM, Du Prez FE. Taking dynamic covalent chemistry out of the lab and into reprocessable industrial thermosets. *Nat Rev Chem* 2025;9: 144–58. <https://doi.org/10.1038/s41570-025-00686-7>.
- [185] Beylergil B, Ozturkmen MB, Al-Nadhari A, Yildiz S, Aydoğan B, Yildiz M. Enhancement of mechanical properties of carbon fiber epoxy composites using methylmethacrylate-butadiene-styrene (MBS) core-shell nanoparticles. *J Compos Mater* 2025. <https://doi.org/10.1177/00219983251315533>.
- [186] Xu H, Chen W-D, Lou Y-P, Xu S-Q, Bai T, Dai W-J, et al. Recent advances in healable carbon-fiber-reinforced epoxy resin composite laminate: preparation and properties. *ACS Appl Polym Mater* 2025. <https://doi.org/10.1021/acsapm.4c03467>.